

A Robust Pixel-Level Interpretability in OCT-based Retinal Disease Classification

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CERTIFICATE

This is to certify that the dissertation entitled “**A Robust Pixel-Level Interpretability in OCT-based Retinal Disease Classification**” submitted by **Mr. Asgar Rashid (12600-BJB-2021)** and **Mr. Faisal Ahmad Malik (65275-ANG-2022)**, to the Department of Computer Applications, Shaheed Himayun Muza-mmil Memorial Government Degree College Anantnag, Higher Education Department, Jammu and Kashmir, India, in partial fulfillment of the requirements for the award of the 4-Year Undergraduate Degree (Honours with Research) has been carried out by the student(s) under my supervision and guidance.

To the best of my knowledge, the work embodied in this dissertation is original and has not been submitted elsewhere for the award of any other degree or diploma.

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DECLARATION

I/we hereby declare that the experimental research work and interpretation of the dissertation entitled “**A Robust Pixel-Level Interpretability in OCT-based Retinal Disease Classification**” as part thereof has not been submitted for any other degree or diploma of any University, nor have the data been derived from any dissertation, thesis or publication of any University or scientific organization. The sources of materials used and all assistance received during the course of investigation have been duly acknowledged.

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ABSTRACT

Retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen are among the leading causes of visual impairment worldwide. Optical Coherence Tomography (OCT) has become the preferred imaging modality for diagnosing these conditions because it provides high-resolution cross-sectional visualization of retinal structures. In recent years, deep learning has demonstrated remarkable success in automatically classifying retinal diseases from OCT images, achieving performance comparable to that of experienced clinicians. However, despite their high predictive accuracy, these models operate largely as black boxes, providing little insight into the reasoning behind their predictions. The absence of transparent and reliable explanations limits clinical confidence and remains one of the major barriers to the widespread adoption of Artificial Intelligence (AI) in ophthalmology. Explainable Artificial Intelligence (XAI) has emerged as a promising solution to this challenge by generating visual explanations that identify the image regions influencing a model's prediction. Nevertheless, existing explainability methods continue to suffer from several important limitations, including coarse localization of pathological regions, limited robustness to image perturbations, and the absence of standardized quantitative evaluation methods for assessing explanation quality. Consequently, although classification performance has improved considerably, the reliability and clinical trustworthiness of generated explanations remain insufficiently addressed. To understand these challenges comprehensively, this research study first presents a detailed background study on retinal anatomy, retinal diseases, Optical Coherence Tomography, deep learning, and Explainable Artificial Intelligence. A systematic literature review and a comparative analysis of representative studies are then conducted to examine the current state of research, identify existing strengths and limitations, and establish the research gaps that motivate this work. The analysis reveals that while considerable progress has been achieved in automated retinal disease classification, comparatively limited attention has been devoted to improving the precision, robustness, and quantitative validation of explainability methods. Motivated by these findings, this research study proposes A Robust Pixel-Level Interpretability in OCT-based Retinal Disease Classification, a framework designed to generate more informative and reliable visual explanations for deep learning-based retinal disease diagnosis. The proposed framework employs an adaptive multi-layer fused Grad-CAM strategy to integrate complementary information from multiple convolutional layers, thereby produc-

ing explanations with improved localization compared to conventional single-layer Grad-CAM. Furthermore, the robustness of the generated explanations is quantitatively evaluated using the Structural Similarity Index Measure (SSIM), providing an objective mechanism for assessing explanation stability under image perturbations. Experimental evaluation demonstrates that the proposed framework generates more consistent and robust explanations while preserving clinically relevant retinal structures. The multi-layer fusion strategy improves the spatial quality of the generated heatmaps, and the quantitative evaluation provides additional confidence in their reliability compared with conventional explainability approaches. By combining improved interpretability with objective robustness assessment, the proposed framework represents a step towards developing more transparent, trustworthy, and clinically applicable AI-assisted diagnostic systems for ophthalmology. Overall, this research study contributes both a comprehensive analysis of existing explainability research in retinal disease classification and a robust pixel-level interpretability framework that addresses important limitations of current approaches. The findings establish a foundation for future research on clinically reliable explainable AI and support the broader integration of trustworthy deep learning systems into ophthalmic diagnosis.

CHAPTER 1

OUTLINE OF THE STUDY

1.1 Introduction

The retina, a complex neural tissue responsible for visual signal transduction, is highly susceptible to degenerative and metabolic pathologies that represent a leading cause of global visual impairment. Conditions such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen-associated Age-Related Macular Degeneration (AMD) involve profound morphological changes to retinal layers. Optical Coherence Tomography (OCT) has emerged as the gold-standard non-invasive imaging modality for diagnosing these diseases, providing micrometer-scale, cross-sectional visualization of the retinal architecture. The manual interpretation of these volumetric datasets is, however, labor-intensive, subject to inter-observer variability, and potentially prone to human error, creating an urgent need for automated diagnostic intervention.

In response, deep learning architectures specifically Convolutional Neural Networks (CNNs) have demonstrated remarkable success in the automated classification of OCT imagery. By autonomously extracting hierarchical, non-linear feature representations, these models routinely achieve classification accuracies that rival or exceed the diagnostic capabilities of general ophthalmologists. However, the deployment of these systems into clinical practice faces a significant paradox. While the classification performance is quantitatively superior, the underlying logic utilized by the neural network remains largely opaque.

These models function as black boxes, where the mapping from raw pixel intensity to a diagnostic output is opaque to the clinician. In medical settings, a binary diagnostic label is insufficient. Ophthalmic treatment protocols, including the administration of intravitreal injections, require explicit, anatomically verifiable justification. The absence of a transparent link between the identified pathology (e.g., a specific cyst or fluid pocket) and the model's prediction undermines clinical trust, regulatory compliance, and medicolegal accountability. This dissertation addresses this critical juncture, focusing on the development and validation of robust, pixel-level interpretability frameworks for retinal disease diagnosis.

1.2 Existing Systems

Current diagnostic systems in ophthalmology have transitioned from traditional machine learning approaches to sophisticated deep learning paradigms. Early automated systems relied heavily on manual feature engineering, utilizing mathematical descriptors such as Local Binary Patterns, Histogram of Oriented Gradients, and Gabor filters to characterize retinal texture. While functional, these methods lacked the hierarchical capacity to model complex disease topologies and were highly sensitive to the intrinsic speckle noise of OCT sensors.

The paradigm shift toward autonomous representation learning enabled by CNNs facilitated significant performance gains. State-of-the-art architectures, including ResNet, VGG, Inception, and EfficientNet, have been extensively validated across various public datasets. These models employ layered convolutional kernels that serve as automatic edge, texture, and structural detectors, enabling them to distinguish between multiple retinal pathologies without the need for manual feature intervention. Furthermore, the recent integration of hybrid architectures fusing CNNs with Transformer-based global attention mechanisms has begun to address the historical limitation of locality in convolution, allowing models to consider global structural dependencies across the entire macula.

To combat the opacity of these models, the field has increasingly adopted Explainable Artificial Intelligence (XAI) as a post-hoc diagnostic tool. Gradient-weighted Class Activation Mapping (Grad-CAM) has become the de-facto standard for spatial interpretability. By backpropagating the classification score to the final convolutional layer, Grad-CAM generates heatmaps that overlay areas of highest influence on the original OCT input. These visualizations attempt to offer a window into the model's internal decision-making, ostensibly highlighting biomarkers such as vascular leakage in CNV or cystoid edema in DME. Despite the widespread adoption of Grad-CAM in modern literature, existing systems remain structurally flawed. They predominantly focus on the terminal layer, leading to coarse, granular heatmaps that blur lesion boundaries and lack the resolution required for high-precision clinical tasks. Furthermore, these explanations have been shown to be mathematically fragile; minor input noise can cause the saliency maps to shift radically, indicating that current XAI methods often prioritize gradient-based artifacts over true clinical pathology.

1.3 Problem Statement

Optical Coherence Tomography is one of the most important imaging modalities available for the diagnosis and longitudinal monitoring of retinal disease, offering clinicians a non-invasive, cross-sectional view of retinal microstructure at near-histological resolution. Over the past decade, deep learning has substantially improved the automated classification of OCT imagery, with convolutional neural networks now routinely matching or exceeding the diagnostic performance of trained ophthalmologists across a range of pathologies. This gain in classification accuracy has, however, come at the cost of interpretability. The internal decision logic of these networks remains largely opaque, and the models are widely characterized as black boxes that provide a diagnostic label without an accompanying, verifiable justification.

This opacity is not merely a theoretical concern. Clinical deployment of any diagnostic system demands a level of trust that a bare classification score cannot supply, particularly in ophthalmology, where treatment decisions such as intravitreal injections or surgical intervention rely on the clinician being able to identify and verify the specific pathological feature driving a diagnosis. Consequently, Explainable Artificial Intelligence has emerged as a necessary companion to deep learning-based diagnostic pipelines, with the goal of rendering the internal reasoning of otherwise opaque networks visible and clinically meaningful.

Among the available XAI techniques, Gradient-weighted Class Activation Mapping and its variants have become the de-facto standard for visual explanation in medical imaging, owing to their architectural simplicity and their ability to generate a spatial heatmap without requiring any modification to the underlying classification network. However, the manner in which these techniques are conventionally implemented introduces a structural constraint: the explanation is derived predominantly from the activations of the last convolutional layer of the network. Because this terminal layer has typically undergone extensive spatial downsampling through repeated pooling and strided convolution operations, the resulting explanation inherits a correspondingly coarse spatial resolution. This dependency on a single, deep layer represents a recurring and largely unaddressed limitation across the reviewed literature, one that carries direct consequences for the quality, boundary precision, and reliability of the explanations produced. It is this limitation, arising from the field's continued reliance on last-layer attribution, that constitutes the research problem addressed by this dissertation.

1.4 Research Gap

Despite the proliferation of diagnostic models, four specific gaps remain unaddressed in the literature:

- Standard Grad-CAM implementations rely on the final convolutional layer of CNNs, which results in extreme spatial downsampling and overly diffuse, low-resolution heatmaps that fail to resolve the precise boundaries of microscopic retinal biomarkers [1, 2].
- Quantitative evaluation frameworks for XAI are currently fragmented and contradictory, lacking a standardized, scientifically coherent methodology for validation [3].
- Current XAI methods demonstrate extreme sensitivity to input noise. Minor changes can lead to the noise problem where heatmaps highlight random pixel clusters rather than true structural pathology [3].

- Most existing works ignore the fact that intermediate CNN layers contain highly relevant spatial detail that is discarded during the forward pass to the final classification layer [4].

1.5 Motivation & Significance

Retinal diseases such as Age-Related Macular Degeneration, Diabetic Macular Edema, and Choroidal Neovascularization collectively affect millions of people worldwide and remain among the leading causes of preventable vision loss. Optical Coherence Tomography is extensively used as the primary imaging modality for detecting and staging these conditions, owing to its ability to resolve retinal microstructure non-invasively and at high spatial resolution. Deep learning has, in turn, greatly improved the automated detection of these pathologies from OCT imagery, reducing the burden of manual review and enabling screening at a scale that would otherwise be impractical for human graders alone. Despite this progress, clinical adoption of deep learning-based diagnostic tools remains constrained. A central obstacle is the lack of transparency inherent to these models: clinicians are understandably reluctant to base treatment decisions on a prediction they cannot independently verify. Existing Explainable AI methods were introduced to address this gap, but, as established in the preceding sections, these methods carry their own limitations, particularly with respect to spatial resolution and boundary precision. This tension between the diagnostic power of deep learning and the practical requirement for verifiable, trustworthy explanations naturally motivates the present study, which is concerned with the systematic literature review and comparative evaluation of interpretability techniques for OCT-based retinal disease classification.

The significance of this study lies in its contribution to improving the understanding and adoption of Explainable Artificial Intelligence (XAI) in OCT-based retinal disease diagnosis. By systematically reviewing and comparing existing research, this study provides ophthalmologists, researchers, and developers with a comprehensive overview of the current approaches used for retinal disease classification and explainability. Such a consolidated analysis enables clinicians to better understand the role of explainable AI in supporting diagnostic decision-making and increasing confidence in AI-assisted healthcare systems. Furthermore, this study identifies the strengths, limitations, and current research trends in OCT-based explainability techniques, providing valuable guidance for researchers interested in developing more reliable and clinically applicable diagnostic models. The findings may also assist healthcare practitioners in understanding the potential of explainable AI to improve transparency, facilitate human–AI collaboration, and support

more informed clinical decisions. Overall, this research serves as a useful reference for future studies aimed at developing trustworthy and clinically acceptable AI systems for ophthalmology.

1.6 Objectives

The specific objectives of this study are as follows:

1. To perform a thorough literature review of the current state of the art.
2. To perform a comparative analysis of OCT classification and Explainable AI techniques in the current state of the art.
3. To design and develop a robust XAI model for OCT classification and interpretability.
4. To test and evaluate the developed model.

1.7 Conclusion

Optical Coherence Tomography occupies a central place in modern ophthalmic practice, providing the high-resolution, cross-sectional visualization of retinal microstructure on which the diagnosis and monitoring of conditions such as Age-Related Macular Degeneration, Diabetic Macular Edema, and Choroidal Neovascularization increasingly depend. Deep learning has substantially advanced the automated interpretation of this imagery, with convolutional and, more recently, transformer-based architectures achieving classification accuracies that rival expert ophthalmologists across a range of pathologies. This capability, however, has not translated directly into clinical trust, since the internal reasoning of these networks remains largely inaccessible to the clinicians who must act on their predictions. Explainable AI has consequently assumed growing importance as the mechanism through which the outputs of otherwise opaque diagnostic models can be rendered interpretable and clinically actionable. Techniques such as Gradient-weighted Class Activation Mapping have been widely adopted for this purpose, owing to their simplicity and compatibility with existing classification architectures. The current explainability methods remain limited in several respects, including coarse spatial resolution, sensitivity to input perturbation, and inconsistent quantitative evaluation practices. These limitations are not isolated to individual studies but recur across a substantial portion of the published work, suggesting a structural gap in how interpretability has thus far been approached in this domain. Before any improved methodology can be meaningfully proposed, it is first necessary to establish, through a systematic and comprehensive literature review, precisely where the current state of the art succeeds and where it falls short.

CHAPTER 2

BACKGROUND AND LITERATURE REVIEW

2.1 Human Retina and Visual System

The human eye is a highly specialized sensory organ responsible for capturing light and converting it into neural signals that are interpreted by the brain as vision. It functions similarly to an optical imaging system, where the cornea and crystalline lens focus incoming light onto the retina, while the iris regulates the amount of light entering through the pupil. The eye works in coordination with the optic nerve and visual cortex to produce clear and accurate perception of the surrounding environment.

2.1.1 Introduction: The Eye as an Optical and Neural Transducer

Vision depends on a biomechanical optical system working together with a finely layered neural processor. At its core, the eye is a biological transducer built to capture electromagnetic radiation in the narrow visible band (roughly 400 to 700 nanometers) and turn those incoming photons into electrochemical signals. This requires the front portion of the eye to remain optically clear while the back portion is kept under tight metabolic control.

Light entering the eye first meets the cornea and the crystalline lens, together known as the dioptric apparatus. The cornea supplies about two-thirds of the eye's total refractive power (around 40 to 44 diopters), thanks to the sharp difference in refractive index between air and the anterior tear film. Since the cornea's shape is fixed, fine focusing, or accommodation, falls to the crystalline lens. Held in place by the zonules of Zinn and shaped by the ciliary muscle, the lens adjusts its curvature so that incoming light converges precisely on the macula. This entire front pathway has to stay free of blood vessels to avoid scattering light, so oxygen and glucose instead arrive via the continuously circulating aqueous humor.

After passing through the vitreous body, a gel-like mixture of water, hyaluronic acid, and a thin collagen scaffold, light reaches the neural retina. Developmentally, the retina isn't really a separate organ at all; it grows out from the diencephalon and is essentially part of the central nervous system. That matters because the retina doesn't behave like passive film. It carries out early stages of compressing, filtering, and sharpening visual information before sending it onward through the optic nerve.

The retina's structure and metabolism both reflect this computational role. To keep optical distortion to a minimum, the unmyelinated axons of the inner retina converge at the optic disc to form the optic nerve. Because myelin scatters light heavily, these axons only gain their myelin sheath after passing the lamina cribrosa, which keeps the inner retinal layers transparent [5].

At the same time, phototransduction is metabolically expensive, so the retina needs a substantial blood supply. It runs on two separate circulations. The inner

two-thirds, from the inner limiting membrane down to the inner nuclear layer, is fed by the central retinal artery through tightly regulated, non-leaky capillaries. The outer third, where the photoreceptors do their demanding work, has no blood vessels at all, since any vasculature there would interfere with incoming light. Instead it gets its nutrients by diffusion from the choroid below, passing through the retinal pigment epithelium (RPE) [6].

This careful separation between vascular supply and neural tissue makes the system fragile. Any structural problem, whether it's drusen building up beneath the RPE [7], mechanical rupture of the RPE-Bruch's membrane complex [8], or breakdown of the inner blood-retinal barrier from chronic high blood sugar [9], can quickly disrupt both the optical and metabolic balance of the tissue. Understanding this basic optical-neural setup is therefore essential before interpreting the disease markers seen on modern optical coherence tomography.

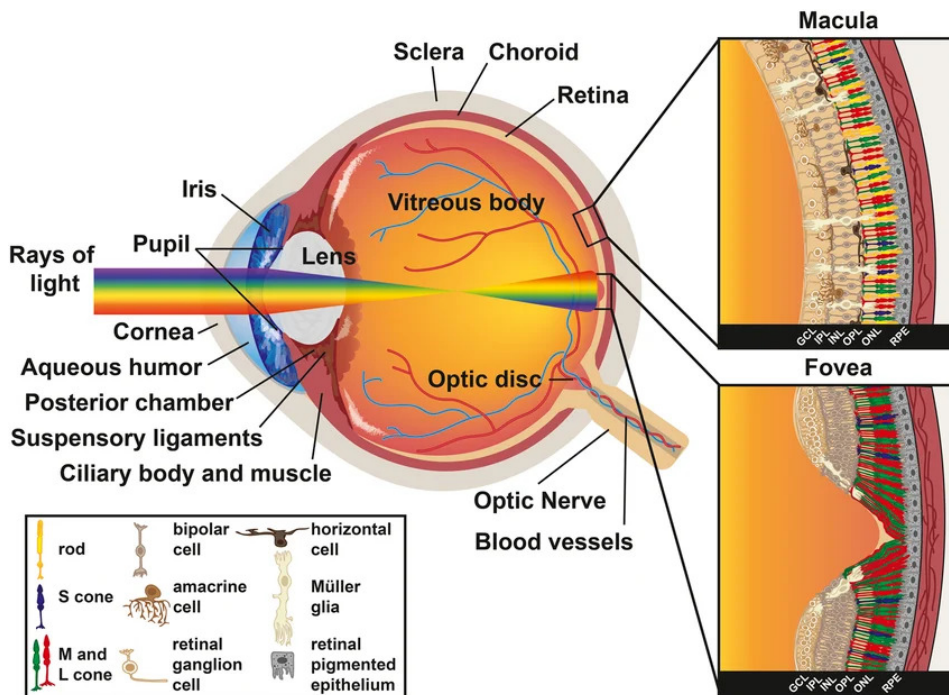


Figure 2.1: Functional schematic of the human eye

2.1.2 Gross Anatomy of the Ocular Globe

The eyeball works as a pressurized, mechanically stable optical instrument that focuses light onto the retina. It isn't a perfect sphere; instead, it's made up of two mismatched curves joined together, a smaller, more steeply curved cornea (radius of curvature about 7.8 mm) and a larger scleral shell (radius about 12 mm). Intraocular pressure (IOP), normally between 10 and 21 mmHg, keeps both segments inflated. This pressure is what keeps the distance between the lens system and the back of

the eye stable, preventing optical errors. To hold this pressure and separate different physiological jobs, the wall of the eye is organized into three concentric layers: an outer fibrous tunic, a middle vascular tunic, and an inner neural tunic.

The fibrous tunic forms the eye's outer shell and is made up of the opaque sclera at the back and the transparent cornea at the front. The sclera is a tough, viscoelastic tissue built mainly from densely packed type I and type III collagen fibers interwoven with elastin. This gives it the strength to resist deformation from both ordinary IOP fluctuations and the pulling forces of the eye muscles. The sclera isn't completely static, though; if its remodeling goes wrong mechanically or biochemically, the eye can elongate abnormally. This underlies high myopia and can lead to severe complications such as scleral staphyloma and rupture of the inner tissue layers, setting the stage for myopic choroidal neovascularization [8].

At the limbus, the sclera gives way to the cornea. Although continuous with the sclera, the cornea achieves transparency through a very precise, crystal-like arrangement of collagen fibers in its stroma. Because even small disruptions to this spacing scatter light, the cornea has to stay in a state of relative dehydration, called deturgescence. This is actively managed by the corneal endothelium, a single layer of hexagonal cells equipped with Na^+/K^+ ATPase pumps that constantly push water out of the stroma and into the anterior chamber against the concentration gradient.

Beneath the fibrous layer sits the uveal tract, or vascular tunic, made up of the iris, ciliary body, and choroid. The front part of the uvea controls how much light enters the eye and manages fluid dynamics in the anterior segment. The non-pigmented ciliary epithelium of the pars plicata secretes aqueous humor, supplying the avascular cornea and lens while also setting baseline IOP.

Behind the ciliary body, the uvea becomes the choroid, a heavily pigmented and richly vascularized tissue layer just outside the retina. The choroid is the main metabolic supply line for the outer eye structures, receiving roughly 85% of total ocular blood flow as a high-volume, low-extraction system. Unlike the brain or other body tissues that try to extract as much oxygen as possible from blood, the choroid's rapid flow mainly works as a heat sink. Light focused onto the macula generates localized heat, and the fast-moving choroidal blood carries that heat away, preventing damage to nearby proteins from overheating.

Looking more closely at its structure, the choroidal blood supply is organized into layers, with vessel size decreasing from outside to inside. The outermost layer, Haller's layer, contains large, non-fenestrated vessels that distribute blood broadly across the back of the eye. Sattler's layer, just inside that, contains medium-sized arterioles that adjust regional blood flow. The innermost layer, the choriocapillaris,

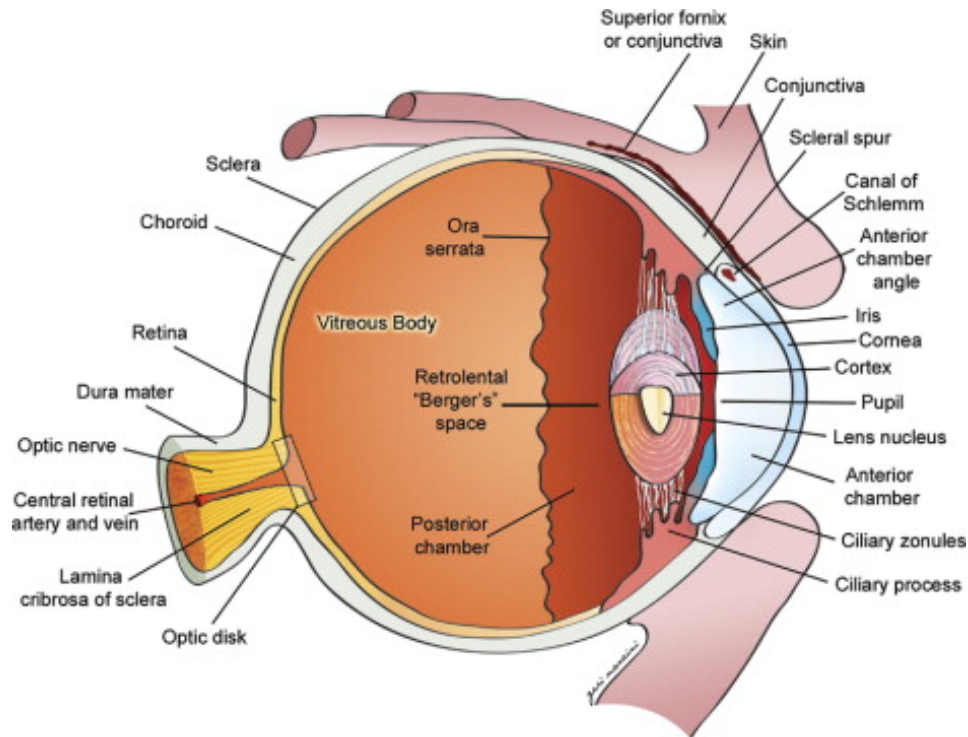


Figure 2.2: Details of the anterior and posterior segments of an eye.

is a dense, flat network of highly interconnected, fenestrated capillaries.

The choriocapillaris is specifically built to feed the metabolically demanding, blood-vessel-free outer retina. Its fenestrations, about 60 to 80 nm across, face almost entirely inward toward the retina, allowing glucose, oxygen, and vitamin A to diffuse quickly across Bruch's membrane. Because the neural retina depends completely on this diffusion route, the health of the choriocapillaris and Bruch's membrane is critical. When lipid and protein debris build up at this interface as drusen, the diffusion distance increases, causing localized ischemia and starting the slow progression toward age-related macular degeneration [7]. Inflammation that disrupts this choroidal architecture can also let new, leaky blood vessels push into the subretinal space, bypassing the normal mechanical barriers [10].

The innermost layer, the neural tunic, is the retina itself. Since it develops as an outgrowth of the diencephalon, the retina is essentially a direct extension of the central nervous system. Its job is to capture incoming photons, carry out early spatial and temporal processing of the visual signal, and send the result onward through the optic nerve as electrochemical impulses. The heavy metabolic demands of phototransduction shape both the retina's layered structure and its dependence on the uveal tissues beneath it.

2.1.3 Laminar Ultrastructure of the Retina

The neural retina has a consistent, layered structure made up of ten distinct histological strata. This arrangement physically separates the biochemical steps of phototransduction from the synaptic networks that perform early visual processing. Since the light-sensitive photoreceptors sit at the very back of the retina, incoming photons have to pass through the transparent, unmyelinated inner layers before they can be captured.

The innermost boundary, facing the vitreous, is the Inner Limiting Membrane (ILM). The ILM is an acellular basal lamina built mainly by the expanded foot-plates of Müller glial cells. It acts as a physical and electrical barrier, resisting the mechanical pulling forces exerted by the vitreous gel during eye movement.

Just outside the ILM is the Retinal Nerve Fiber Layer (RNFL), made up of the unmyelinated axons of the retinal ganglion cells (RGCs), which travel in distinct arc-shaped bundles toward the optic nerve head. Because myelin would scatter light and reduce visual clarity, these fibers only gain their oligodendroglial myelin sheath after passing through the lamina cribrosa to form the post-laminar optic nerve [5].

Below the RNFL sits the Ganglion Cell Layer (GCL), which contains the cell bodies of the RGCs along with a small number of displaced amacrine cells. How densely packed this layer is varies a great deal across the retina. In the periphery, the GCL is a single, sparsely populated layer, while in the central macula, optimized for high-resolution vision, it can stack up to six cell layers deep to handle the processing load from the tightly packed foveal cone mosaic [5].

The Inner Plexiform Layer (IPL) is where most of the inner retina's synaptic activity happens. Here, the dendrites of RGCs form excitatory synapses with bipolar cell axon terminals, and this signal transfer is continuously shaped by lateral inhibition from amacrine cells, which adjust the temporal and spatial contrast of the visual signal before it heads to the visual cortex [5].

In clinical practice, especially with structural OCT, the RNFL, GCL, and IPL are often grouped together into one measurement called the Ganglion Cell Complex (GCC). The GCC covers the full extent of the RGCs, from dendrites to axons. Thinning of the GCC is a sensitive sign of irreversible inner retinal neurodegeneration and forms the basis for monitoring optic neuropathies [5, 6].

The Inner Nuclear Layer (INL) sits at the center of the retina and contains the densely packed nuclei of bipolar, horizontal, and amacrine cells, along with the cell bodies of Müller glia, which span the retina vertically. Bipolar cells act as the main vertical relay, passing signals from photoreceptors to RGCs. Müller cells run from the ILM to the photoreceptor layer and serve as the retina's main housekeepers,

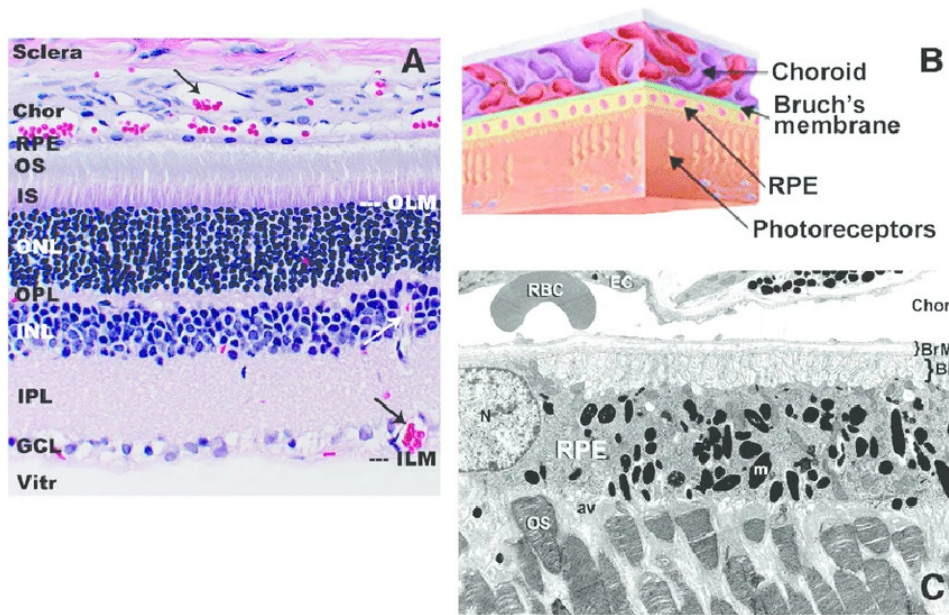


Figure 2.3: Human retinal cross-section illustrating the ten retinal layers and the Ganglion Cell Complex (GCC).

using aquaporins and inward-rectifying potassium channels (Kir4.1) to drain excess fluid and regulate potassium levels, preventing fluid buildup within the retina [9].

The Outer Plexiform Layer (OPL) is where photoreceptor terminals, the rod spherules and cone pedicles, connect with bipolar and horizontal cell dendrites. Horizontal cells provide strong lateral inhibition here, creating the center-surround response that's important for edge detection. In the fovea, the inner layers shift sideways, forcing photoreceptor axons to run obliquely to keep their connections intact. This oblique bundle of axons is known as Henle's fiber layer and shows distinctive birefringence on directional OCT [9].

The Outer Nuclear Layer (ONL) contains only the cell bodies of rods and cones. Its thickness tracks how densely packed the photoreceptors are, reaching its peak in the foveola where cone bodies are crammed together to maximize sharpness.

Beyond the ONL is the External Limiting Membrane (ELM). Despite its name, the ELM isn't a true membrane but rather a row of junctional complexes (zonula adherens) anchoring the apical processes of Müller cells to the photoreceptor inner segments. It acts as a sieve, blocking large proteins from entering the subretinal space while keeping the photoreceptors properly aligned [9, 11].

The Photoreceptor Inner and Outer Segments (IS/OS) form the metabolic and sensory core of vision. The inner segments are packed with mitochondria to meet the enormous energy demands of phototransduction and the visual cycle. On high-resolution OCT, this dense mitochondrial packing shows up as a bright band known clinically as the ellipsoid zone (EZ). The outer segments contain stacked membrane

discs loaded with opsin pigments, which physically capture photons and kick off the G-protein-coupled phototransduction cascade [11, 12].

The outermost layer of the retina is the Retinal Pigment Epithelium (RPE), a single layer of pigmented, hexagonal cells firmly attached to Bruch's membrane. The RPE performs several essential jobs. Its tight junctions form the outer blood-retinal barrier, controlling the flow of glucose, oxygen, and vitamin A from the choriocapillaris to the avascular outer retina. It also acts as a dark heat sink, with melanin granules absorbing stray light and reducing oxidative damage. Metabolically, the RPE continuously recycles shed photoreceptor outer segment discs through phagocytosis, returning photopigments to the visual cycle and preventing toxic buildup of oxidized lipids [11, 12].

2.1.4 Cellular Physiology and Phototransduction

Phototransduction is the process that turns captured light into an electrochemical signal, and it happens entirely within the outer segments of rods and cones. While both cell types share the same basic machinery, they use different opsins and have different shapes that suit their roles. Rods, used for low-light vision, are packed with free-floating disc membranes loaded with rhodopsin, giving them the sensitivity to detect a single photon. Cones, used for daylight and color vision, have continuous, deeply folded membranes with photopsins tuned for fast responses and color discrimination [12].

A photoreceptor's resting state is somewhat unusual compared to other neurons. In darkness, it stays continuously depolarized due to what's called the "dark current." High levels of cyclic guanosine monophosphate (cGMP) keep cyclic nucleotide-gated (CNG) channels in the outer segment open, allowing a steady influx of sodium and calcium ions. To balance this, the inner segment runs Na⁺/K⁺ pumps continuously to push these ions back out [9]. This constant cycling holds the resting membrane potential around -40 mV, keeping voltage-gated calcium channels at the synaptic terminal open and maintaining a steady release of the inhibitory neurotransmitter glutamate into the outer plexiform layer [5].

When a photon hits the outer segment and is absorbed, things change quickly. Rhodopsin, the classic photopigment, is a G-protein-coupled receptor made of an opsin protein bound to 11-*cis*-retinal. Absorbing a photon causes this molecule to switch shape, isomerizing into all-*trans*-retinal. This forces the surrounding opsin into its active form, Metarhodopsin II.

Metarhodopsin II then triggers a large signal-amplifying cascade. Each activated Metarhodopsin II molecule activates hundreds of transducin (G_t) molecules. Transducin's alpha subunit swaps GDP for GTP, and the activated complex sepa-

rates and binds to phosphodiesterase 6 (PDE6), relieving its inhibition. PDE6 then rapidly breaks down cGMP into 5'-GMP, thousands of molecules per second.

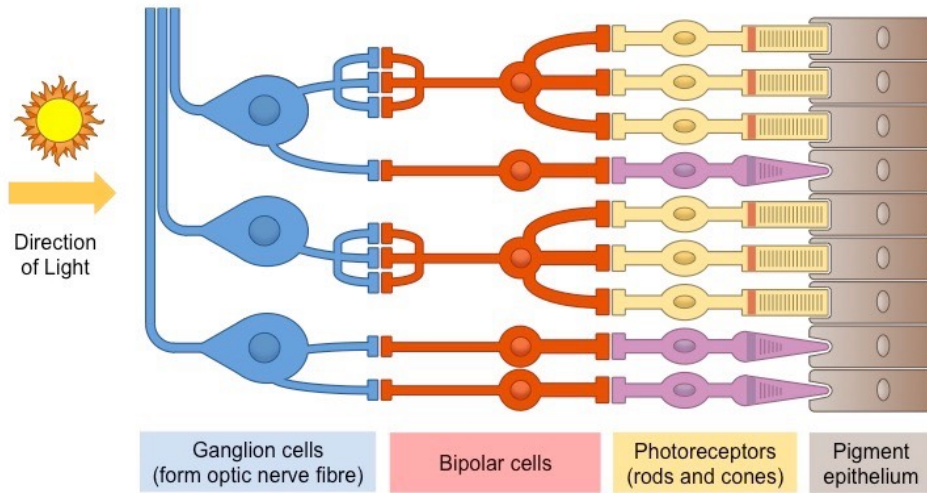


Figure 2.4: Biochemical schematic of the phototransduction cascade.

As cGMP levels drop, the CNG channels close. Sodium and calcium influx stops, but the inner segment's pumps keep working, pulling positive charge out of the cell. This shifts the membrane potential from around -40 mV down to roughly -70 mV, a hyperpolarized state.

This hyperpolarization spreads to the synaptic terminal and closes the voltage-gated calcium channels there, stopping glutamate release. So the actual "signal" sent from a photoreceptor isn't a spike, it's a drop in an otherwise steady stream of inhibitory neurotransmitter [5]. This drop in glutamate has opposite effects on the two types of bipolar cells downstream: it excites ON-bipolar cells, which use metabotropic mGluR6 receptors, and inhibits OFF-bipolar cells, which use ionotropic AMPA/kainate receptors, setting up parallel pathways for contrast detection.

To reset and stay responsive, the cascade has to shut down quickly. Rhodopsin kinase and arrestin deactivate Metarhodopsin II. Falling calcium levels activate guanylate cyclase-activating proteins (GCAPs), which trigger guanylate cyclase to make fresh cGMP, reopening the CNG channels and restoring the dark current. Meanwhile, the spent all-*trans*-retinal is released from the opsin, converted to all-*trans*-retinol, and shipped to the RPE. There, isomerohydrolase enzymes convert it back to 11-*cis*-retinal, which returns to the outer segment to regenerate working rhodopsin [7]. This tight metabolic loop explains why problems at the RPE quickly affect photoreceptor health and vision.

2.1.5 The RPE and Bruch's Membrane Complex

The boundary between the richly vascularized uveal tract and the avascular neural retina is governed by the RPE and Bruch's membrane working together. These two structures aren't independent; they function as a single interdependent unit controlling the environment of the outer retina. Since photoreceptors have no blood supply of their own and produce large amounts of oxidative waste, they rely entirely on this complex to move nutrients, fluid, and waste products in both directions.

The RPE is a continuous, single layer of pigmented, hexagonal cells. These cells are strongly polarized: their apical surfaces have dense microvilli that wrap around the photoreceptor outer segments without forming actual junctions with them. On their sides, RPE cells are sealed together by tight junctions near their apical edges, forming the outer blood-retinal barrier (BRB) [9]. This barrier blocks unregulated movement of fluid, large molecules, and inflammatory mediators from the highly fenestrated choriocapillaris into the subretinal space, meaning anything that crosses has to go through the RPE cells themselves.

To keep the retina pressed against the back of the eye, the RPE works as a continuous pump. Using basal chloride channels and Na⁺/K⁺ ATPase pumps, it actively moves water out of the subretinal space and into the choroid against an osmotic gradient. This steady fluid movement creates suction that holds the photoreceptors against the RPE without needing physical attachments. If this pumping fails, or if the tight junctions of the outer BRB break down, fluid quickly accumulates under the retina, separating photoreceptors from their supply and damaging vision.

Just beneath the RPE lies Bruch's membrane, an acellular extracellular matrix that acts as the main filter for the back of the eye. It's about 2 to 4 micrometers thick and consists of five layers: the RPE's basal lamina, an inner collagenous zone, a central elastin layer, an outer collagenous zone, and the basal lamina of the choriocapillaris endothelium [7]. This layered structure gives it the elasticity to handle changes in choroidal blood pressure while letting small nutrients like glucose, oxygen, and vitamin A diffuse inward and waste move outward.

Bruch's membrane gets stiffer and less permeable with age. Ongoing oxidative stress causes its collagen and elastin fibers to cross-link and stiffen. At the same time, lipids and protein debris released by the RPE get trapped within its collagenous layers. When this debris builds up between the RPE basal lamina and the inner collagenous layer, it becomes drusen [7,11]. As drusen grow, they widen the gap between the choriocapillaris and RPE, causing chronic low oxygen in the outer retina and driving the progression of age-related macular degeneration (AMD). If the central elastin layer thins or fractures badly enough, it can open a path for inflammatory

cells and endothelial cells to invade, leading to choroidal neovascularization (CNV) breaking through into the sub-RPE and subretinal spaces [10].

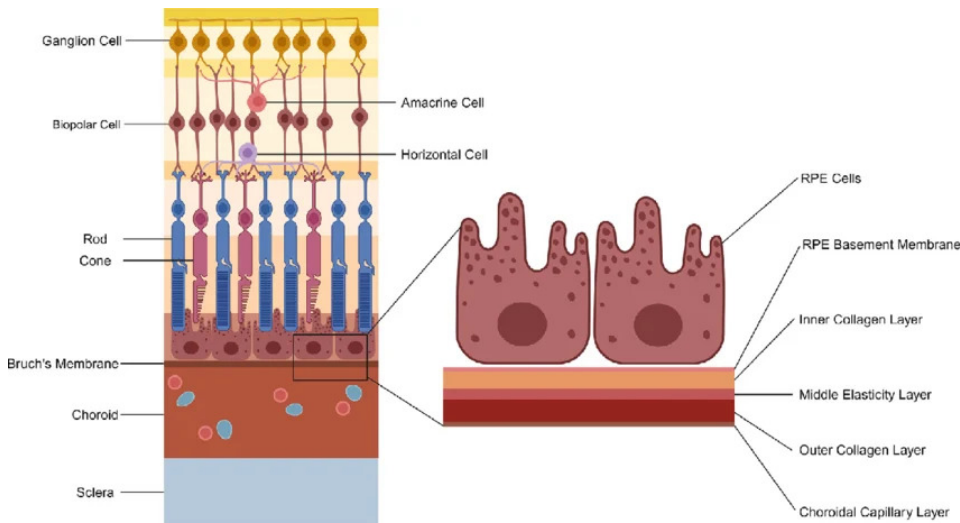


Figure 2.5: Ultrastructural schematic of the RPE–Bruch’s membrane complex

Photoreceptor survival depends heavily on the RPE’s capacity to break down waste. To keep the phototransduction machinery working well, rod and cone outer segments are constantly renewed: new discs form at the base while older, damaged discs are shed from the tip. The RPE engulfs these shed discs and breaks them down using lysosomes [9]. A single RPE cell handles waste from up to forty photoreceptors every day. Over the decades, incomplete breakdown of this material leaves behind lipofuscin, an autofluorescent lipid-protein byproduct. As lipofuscin accumulates inside RPE cells, it crowds the cytoplasm and reduces the cell’s ability to do its job, eventually contributing to geographic atrophy.

The RPE also plays a central role in the visual cycle itself. After a photon is captured, the resulting *all-trans*-retinol needs to leave the photoreceptor quickly to avoid toxicity. Binding proteins shuttle it across the subretinal space into the RPE, where enzymes including lecithin retinol acyltransferase (LRAT) and RPE65 convert it back into *11-cis*-retinal. This regenerated molecule is then sent back to the photoreceptors, keeping the supply of light-sensitive pigment steady.

2.1.6 Topographical Specialization and Blood Supply

The retina isn’t uniform; its structure and blood supply vary across its surface to balance sharp central vision against peripheral motion detection in low light. At the center of vision sits the macula lutea, roughly 5.5 mm across. It’s marked by high concentrations of the pigments lutein and zeaxanthin in the axons of the inner retina, which filter out high-energy blue light, reducing chromatic aberration and protecting the photoreceptor-RPE complex from photo-oxidative stress [7].

At the center of the macula is the fovea centralis, about 1.5 mm across, marked by a distinct dip. To achieve the sharpest possible vision, the fovea pushes the scattering inner layers (RNFL, GCL, IPL, and INL) out of the way, forming the foveal pit, so light hits the densely packed, rod-free cone mosaic directly. Because the secondary neurons are displaced sideways, the axons of foveal cones have to stretch and angle to keep their connections, forming Henle's fiber layer, a structure that's particularly vulnerable to fluid buildup during exudative disease [12].

This demanding tissue needs a correspondingly specialized dual blood supply. The avascular outer retina relies entirely on diffusion from the choroid across Bruch's membrane. The inner two-thirds of the retina is fed directly by the central retinal artery (CRA), a branch of the ophthalmic artery. After emerging from the optic nerve head, the CRA splits into superior and inferior arcades running mainly through the RNFL. Unlike the choroid, retinal circulation is a low-flow, high-extraction system controlled by local autoregulation rather than by the autonomic nervous system.

As these arterioles branch into the retina, they form distinct, depth-specific capillary networks. In the macula, this microvasculature organizes into three layers: the superficial capillary plexus (SCP) in the RNFL and GCL, the intermediate capillary plexus (ICP) at the inner border of the INL, and the deep capillary plexus (DCP) at its outer border [13, 14]. A fourth layer, the radial peripapillary capillary (RPC) network, exists only around the optic disc to support the thick RNFL axon bundles converging there [5].

To protect the optical clarity of the foveola, retinal capillaries stop short of the very center, leaving the Foveal Avascular Zone (FAZ), roughly 0.3 to 0.5 mm across, with no inner retinal blood vessels at all. The tissue here survives purely on diffusion from the choriocapillaris below. Because it sits at the very edge of the retina's perfusion, the FAZ is especially sensitive to ischemia. Enlargement of the FAZ, capillary dropout around it, and loss of its normal round shape are all useful early markers of microvascular damage in diseases like diabetic retinopathy [13, 15].

The endothelial cells lining retinal capillaries are quite different from those in the choroid. They're non-fenestrated and sealed by continuous tight junctions. Together with the basement membrane, pericytes, and the end-feet of astrocytes and Müller cells, they form the neurovascular unit, which maintains the inner blood-retinal barrier (BRB). This barrier tightly controls ion movement and completely blocks plasma proteins and fluid from leaking out between cells. When chronic high blood sugar and the buildup of advanced glycation end-products (AGEs) kill off pericytes, these tight junctions break down, allowing fluid and lipid to leak into

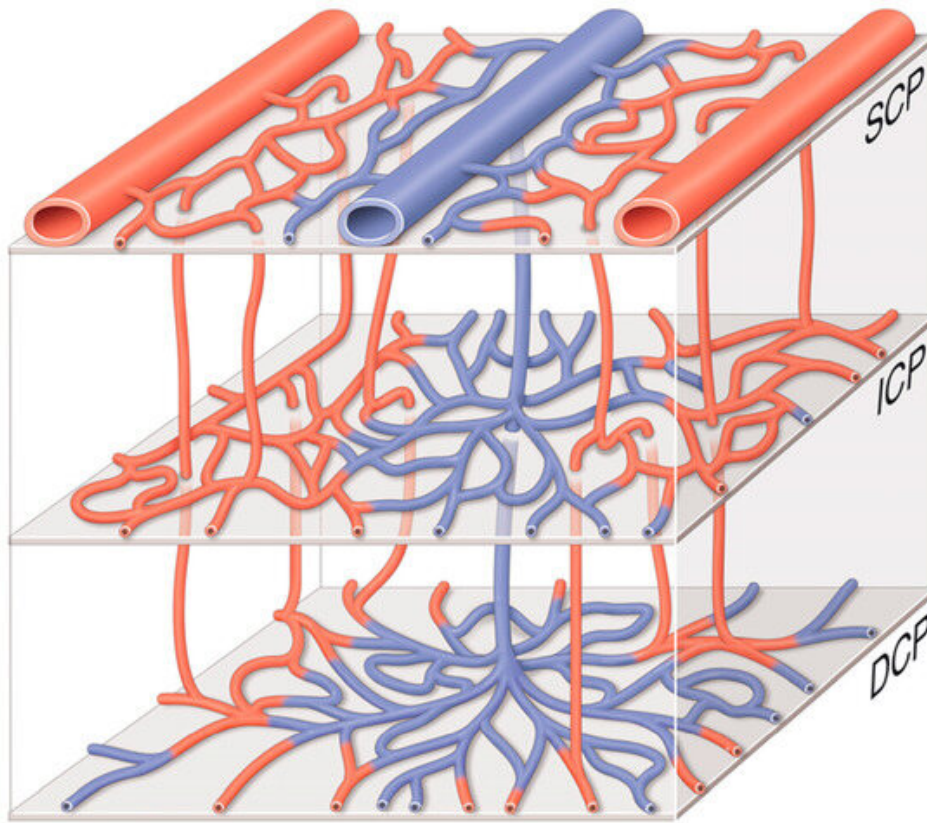


Figure 2.6: Schematic representation of the retinal vascular network.

the retina, the central mechanism behind diabetic macular edema (DME) [9, 16].

2.2 Retinal Pathologies and Morphological Biomarkers

Retinal diseases alter the normal architecture of the retina through a variety of pathological processes that affect its cellular organization, vascular integrity, and supporting tissues. These structural changes manifest as characteristic morphological features that can be visualized using modern retinal imaging techniques. Recognizing these biomarkers is fundamental to disease diagnosis, progression assessment, and the development of automated classification systems.

2.2.1 Choroidal Neovascularization (CNV)

Choroidal neovascularization (CNV) occurs when the outer blood-retinal barrier fails completely, allowing abnormal new blood vessels to grow from the choriocapillaris. These vessels break through Bruch's membrane and the RPE, pushing into the sub-RPE or subretinal space. This physically displaces the photoreceptors above, and because these new vessels are leaky and lack proper tight junctions, they allow lipids, plasma proteins, and blood to spill into the surrounding tissue. The main driver of this abnormal vessel growth is excess Vascular Endothelial Growth Factor A (VEGF-A).

Normally, the RPE secretes a steady low level of VEGF toward the choroid to keep the choriocapillaris healthy and fenestrated. But under hypoxic stress, oxidative damage, or inflammation, the RPE and nearby macrophages shift gears and ramp up VEGF production dramatically, driven by hypoxia-inducible factor 1- α (HIF-1 α). This VEGF binds to VEGFR-2 receptors on choroidal endothelial cells, triggering signaling through PLC γ and PI3K/Akt pathways that pushes these cells to remodel, proliferate, and migrate toward the source of the signal.

For new vessels to actually reach the retina, Bruch's membrane has to be structurally compromised first; its intact layered matrix is normally an effective barrier to invading vessels. Breaking through requires enzymatic degradation by matrix metalloproteinases, particularly MMP-2 and MMP-9, secreted by stressed RPE cells and immune cells. These enzymes break down the collagen and elastin in the membrane's central layers, and once that matrix is sufficiently weakened or torn, choroidal endothelial cells can exploit the gaps and form new vessel networks that bypass the outer BRB entirely.

While the underlying VEGF-driven biology is similar across most types of CNV, what triggers it shapes how the lesion looks, how much fluid it produces, and how it progresses clinically. Three main causes lead to this kind of breakdown: localized inflammation, severe mechanical failure of the eye's structure, and rare anomalies seen in children.

Inflammatory CNV (iCNV) mostly affects younger patients and develops from various non-infectious forms of posterior uveitis, especially punctate inner choroidopathy (PIC) and multifocal choroiditis (MFC) [10,17]. Unlike age-related CNV, iCNV isn't caused by aging or drusen but by acute, localized inflammation. Macrophages, T-cells, and giant cells infiltrate the choroid, forming granulomas that disrupt the RPE and release large amounts of both VEGF and MMPs directly [17]. These lesions effectively dissolve Bruch's membrane and act as direct routes for new vessel growth. Because the surrounding tissue is so inflamed and the local vasculature dilated, iCNV typically presents with intense, rapid fluid leakage and quick vision loss unless treated aggressively with steroids and anti-VEGF therapy [10].

Myopic CNV (mCNV), by contrast, is driven almost entirely by mechanical stretching of the back of the eye. In pathologic myopia, defined by an axial length over 26.5 mm or a refractive error worse than -6.00 diopters, the eye's continued elongation places enormous shear stress on the sclera and choroid [8]. As these tissues thin out, the choriocapillaris deteriorates and blood supply to the outer retina drops. Worse, the elastin and collagen in Bruch's membrane lose their elasticity under this stretching, leading to spontaneous tears known as lacquer cracks [8].

These small tears give nearby choroidal vessels, already weakened by the stretching, an easy path into the subretinal space. As a result, mCNV lesions tend to be small, flat, and produce relatively little fluid compared to other forms. Even so, they're highly destructive: they trigger localized RPE pigmentation and fibrous scarring, called the Fuchs spot, that permanently destroys the foveal photoreceptors beneath it, leaving a permanent blind spot in central vision [8].

A separate and rare category, pediatric CNV, behaves quite differently from adult forms. A large retrospective study of children and adolescents in China found that pediatric CNV is mostly idiopathic, inflammatory, or linked to congenital abnormalities like optic nerve head drusen or choroidal osteomas [18]. Because pediatric RPE and retinal tissue heal much better than aging tissue, pediatric CNV often resolves on its own. These patients typically need far fewer anti-VEGF injections than adults to reach a stable state, though they still need long-term monitoring for recurrence [18].

Putting these pathophysiological changes into measurable terms requires specialized volumetric imaging. On standard cross-sectional OCT, active CNV shows up as a structurally abnormal mass, known as Subretinal Hyperreflective Material (SHRM), located either just below the RPE (Type 1 CNV) or breaking through the RPE to sit beneath the neurosensory retina (Type 2 CNV) [19]. This bright mass corresponds to a mix of new vessels, displaced RPE cells, and fibrin, and it's typically surrounded by darker areas representing fluid pockets, whether subretinal fluid or intraretinal cysts.

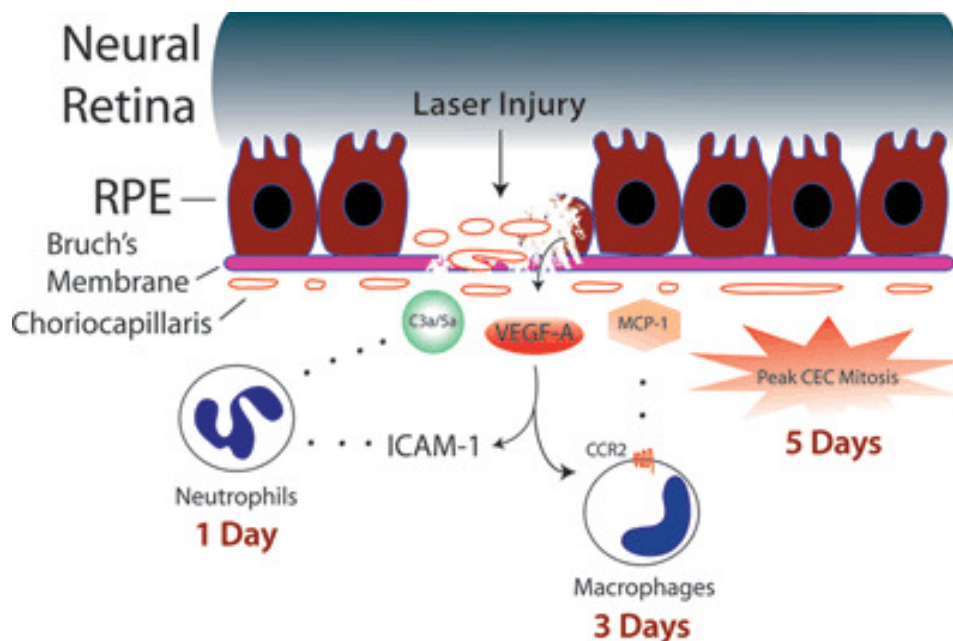


Figure 2.7: Pathogenetic mechanisms of choroidal neovascularization (CNV).

The way CNV is diagnosed changed substantially with the arrival of Optical Coherence Tomography Angiography (OCTA). Standard OCT only shows the static reflectance of tissue, while OCTA picks up motion-contrast signals from flowing red blood cells across repeated B-scans, mapping blood flow directly without injected dye [14]. This gives a direct, three-dimensional view of the entire neovascular network.

On en-face OCTA images, active CNV networks often produce distinctive branching patterns described as "medusa-head" or "sea-fan" shapes [14]. These show a large central feeder vessel passing through the gap in Bruch's membrane, spreading outward into a tightly woven fringe of vessels at the lesion's edge, the active growing front where VEGF-driven vessel growth is concentrated. Around this network, OCTA often shows a dark halo, representing a localized loss of choriocapillaris flow as the new high-flow vessels essentially steal blood from the surrounding normal capillaries, worsening local hypoxia and pushing the disease to spread further [14].

2.2.2 Drusen and Sub-RPE Deposits

The buildup of debris beneath the RPE, known as drusen, is the defining early sign of age-related macular degeneration (AMD). Drusen reflect a chronic local failure of the RPE-Bruch's membrane system. Normally, the RPE processes huge volumes of shed photoreceptor material and manages metabolic traffic in both directions. With age and accumulated oxidative stress, the RPE's ability to break down this material declines. Partly digested lipid-protein complexes get pushed out toward an increasingly stiff and less permeable Bruch's membrane, and because that membrane's permeability drops with age, this material gets trapped between the RPE's basal lamina and the membrane's inner collagenous layer [7].

Chemically, drusen are far from simple; they're complex, layered deposits that reflect the underlying lipid-processing and inflammatory dysfunction. They contain a core rich in cholesterol, both free and esterified, along with apolipoproteins ApoE and ApoB, pointing to the RPE as their main source. Mixed in are immune-related molecules including vitronectin, amyloid-beta, and complement proteins such as Complement Factor H (CFH) and the Membrane Attack Complex (MAC) [7]. These complement components show that drusen aren't just passive physical barriers; they create a low-grade, ongoing inflammatory environment that continually stresses the RPE.

Drusen are also classified by their size and how they appear on OCT, which relates directly to clinical risk. Hard drusen are the earliest form, appearing as small, well-defined yellow spots under $63\text{ }\mu\text{m}$ wide. On OCT, they show up as small, sharply outlined dome-shaped bumps in the RPE with uniform brightness.

Because they're small and isolated, hard drusen barely affect the diffusion gradient and carry little short-term risk.

Soft drusen are more concerning. They spread wider than 125 μm and have poorly defined, sloping edges. Their makeup differs too, containing more loose lipid material, which makes them prone to merging with neighboring drusen. When soft drusen combine, they form large drusenoid pigment epithelial detachments (PEDs) that cut off the RPE and photoreceptors above from the choriocapillaris, causing significant local ischemia. On OCT, soft drusen appear as broad, irregular elevations with moderately reflective, mixed-looking material inside [20].

Cuticular drusen, sometimes called basal laminar drusen, look quite different. Instead of spreading out individually, they form countless small, tightly packed bumps that can cover much of the back of the eye. On OCT, they create a distinctive "sawtooth" pattern along the RPE [12]. Despite their small size, their sheer number changes how the RPE behaves mechanically and can predispose it to sudden vitelliform detachments.

Reticular pseudodrusen (RPD), also called subretinal drusenoid deposits (SDD), are different again, they sit above the RPE rather than below it, in the space between the RPE's microvilli and the external limiting membrane [12]. This location makes them especially harmful, since they directly abrade the photoreceptor outer segments and disrupt the ellipsoid zone, rather than just affecting diffusion. On high-resolution OCT, they appear as bright, cone-shaped mounds pushing into the subretinal space. Their presence is strongly linked to a thin choroid and is an independent risk factor for rapid progression to geographic atrophy and Type 3 macular neovascularization.

Moving from qualitative fundus photos to quantitative OCT has changed how drusen are assessed. Measuring area alone doesn't capture the full picture; what really matters for ischemic risk is total drusen volume. Modern segmentation algorithms can now precisely trace the RPE and Bruch's membrane to measure the apical height and basal width of individual deposits [11,21]. The basal width determines how much of the choroidal diffusion area is blocked, while the apical height reflects how much mechanical stress is placed on the RPE [20].

When the stress from growing drusen becomes too much, the RPE starts to fail. Stressed RPE cells detach from their basal lamina, undergo a transition resembling mesenchymal cells, and migrate into the retina above. Pigment-laden macrophages also move into the inner layers to clean up oxidized lipid debris. On OCT, these migrating cells appear as small, bright spots called Hyperreflective Foci (HRF) [12].

The presence of HRF signals active, ongoing damage and is a reliable warning

sign of disease progression. The functional impact is measurable: microperimetry studies show that retinal sensitivity drops in areas with high drusen volume and HRF density, often well before any cells actually die or geographic atrophy becomes visible [22].

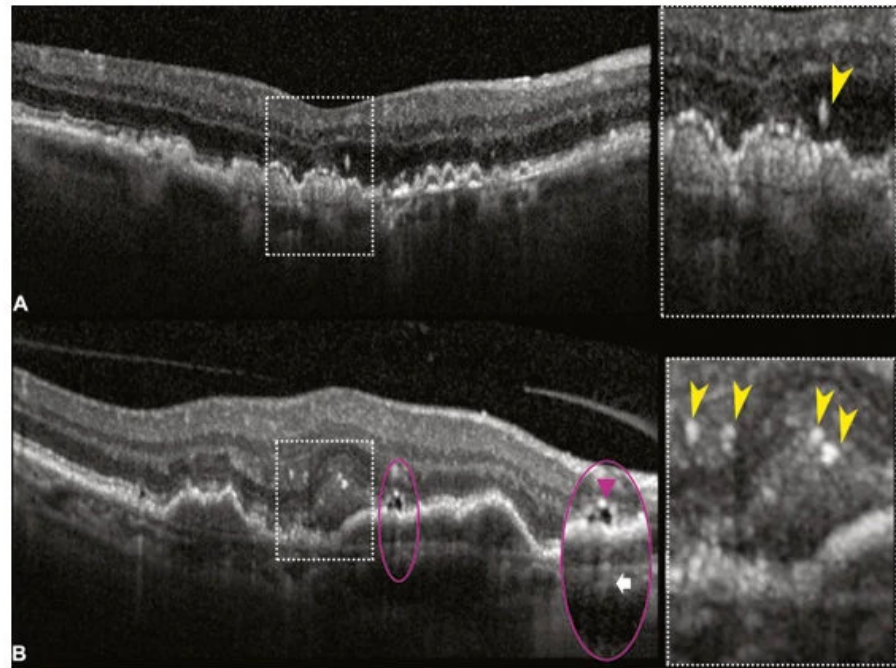


Figure 2.8: Structural OCT B-scans of intermediate age-related macular degeneration (AMD).

2.2.3 Diabetic Macular Edema (DME)

Diabetic Macular Edema (DME) is the leading cause of vision loss in people with both Type 1 and Type 2 diabetes. Unlike the slow degenerative process behind AMD, DME comes from a breakdown of the inner blood-retinal barrier (BRB) driven by microvascular damage and inflammation. This breakdown lets fluid, proteins, and lipids leak directly into the organized tissue of the macula.

DME starts with chronic high blood sugar, which damages the retinal neurovascular unit, made up of non-fenestrated capillary endothelial cells, their basement membrane, pericytes, and the surrounding astrocyte and Müller cell processes. Normally, this unit tightly controls what crosses between blood and tissue, blocking anything from passing between cells thanks to tight junction proteins like claudins, occludins, and zonula occludens-1.

Persistently high intracellular glucose overwhelms normal glucose processing and pushes excess sugar into harmful side pathways. The most damaging of these is non-enzymatic glycation, which produces Advanced Glycation End-products (AGEs).

These build up in the vascular basement membrane, cross-linking proteins and stiffening capillary walls [9,16]. AGEs also bind to RAGE receptors on endothelial cells and pericytes.

This AGE-RAGE binding causes a burst of reactive oxygen species (ROS) via NADPH oxidase activation, which in turn activates NF- κ B and sets off a localized inflammatory response [9]. Endothelial cells then increase expression of ICAM-1, which traps circulating white blood cells (leukostasis). These cells physically clog capillaries, creating small ischemic zones in the macula while also damaging the endothelium directly.

One of the earliest and most damaging events in this process is the death of pericytes. Pericytes normally provide mechanical support and secrete survival signals like angiopoietin-1 that help maintain endothelial tight junctions. When high glucose levels cause pericytes to die off, leaving behind so-called "ghost cells," capillary walls become structurally weak and unstable [16]. Without pericyte support and under ongoing ischemic stress from leukostasis, the surrounding tissue becomes hypoxic. In response, Müller cells and stressed neurons ramp up production of VEGF-A along with inflammatory cytokines such as Interleukin-6 (IL-6) and Tumor Necrosis Factor-alpha (TNF- α) [9, 15].

This wave of cytokines is what finally breaks down the inner BRB. VEGF causes phosphorylation of tight junction proteins between endothelial cells, pulling them apart and opening gaps between cells. At the same time, increased transcytosis allows plasma to move across endothelial cells even where the membrane is intact. Driven by blood pressure from the central retinal artery, fluid and lipids then flow into the inner nuclear layer (INL) and outer plexiform layer (OPL) [13].

How this fluid overload appears structurally depends on how long and how severe the leakage has been, and this is something structural OCT captures very precisely. Early DME often shows up as diffuse, sponge-like swelling, reflecting cellular swelling and low-level fluid buildup between cells. As pressure increases, this progresses to classic Cystoid Macular Edema (CME), where ongoing fluid influx physically tears apart synapses in the INL and OPL. Müller cells, overwhelmed by the osmotic load and with reduced aquaporin-4 and Kir4.1 channel activity, begin to die [9]. The resulting fluid collects in large, dark intraretinal cysts (IRC). In the fovea, these cysts follow the angled axons of Henle's fiber layer, producing the characteristic petal-shaped pattern seen on angiography.

Advanced quantitative OCT analysis has identified several structural biomarkers that predict functional outcomes in DME far better than simple central subfield thickness measurements.

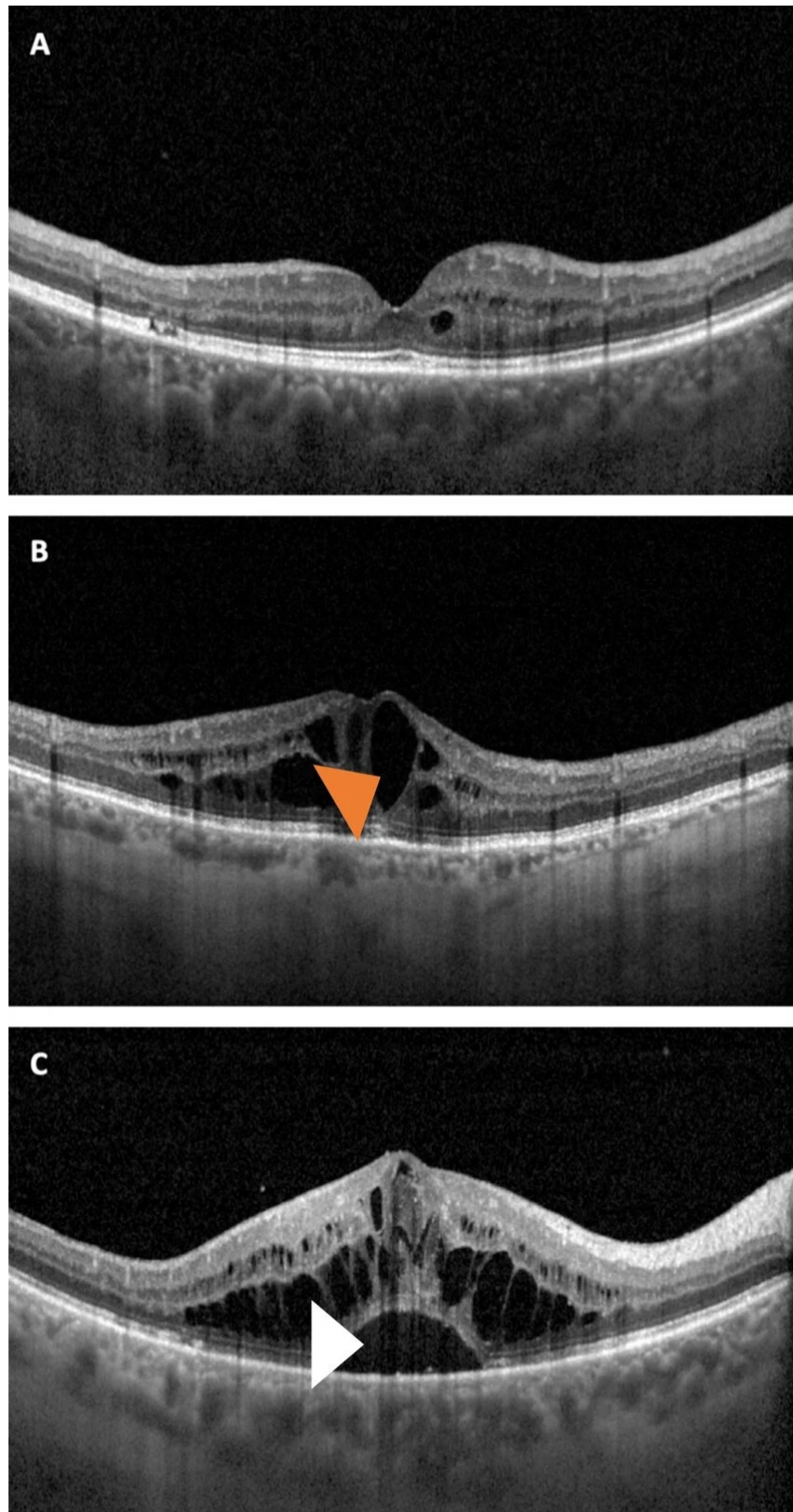


Figure 2.9: Structural OCT B-scans illustrating the morphological spectrum of diabetic macular edema (DME).

2.3 Retinal Imaging Modalities

Advances in retinal imaging have transformed the diagnosis and management of ocular diseases by enabling detailed visualization of retinal anatomy and pathology. Different imaging modalities provide complementary structural and functional information, each offering distinct advantages depending on the clinical application. Together, these technologies form the imaging foundation for both clinical ophthalmology and computer-aided diagnostic systems.

2.3.1 Traditional Two-Dimensional Imaging

The retina and the uveal tissue supporting it form a transparent, layered structure, and clinicians need ways to assess them quantitatively without invasive biopsies. Historically, this meant relying on optical projection methods that captured the back of the eye as flat, two-dimensional images.

The most basic of these is Color Fundus Photography (CFP), which simply relies on backscattered light. The system illuminates the back of the eye with broad-spectrum visible light (400 to 700 nm) and captures the reflection with a CMOS or CCD sensor. The resulting image depends on how different tissues absorb light: melanin in the RPE and choroid absorbs broadly, hemoglobin in blood vessels absorbs green and blue light strongly, and macular pigments filter out blue light selectively [21].

CFP is good at showing high-contrast surface features like the optic disc, the major blood vessel arcades, and large drusen, but it has one fundamental limitation. Because it integrates all the reflected light along the depth axis into a single pixel value, it completely loses depth information. CFP can't measure how tall a drusen deposit is, nor can it tell apart a shallow pocket of subretinal fluid from severe intraretinal cystic swelling [6].

To look at blood flow and the function of the blood-retinal barriers, clinicians turned to fluorescence imaging, primarily Fluorescein Angiography (FA). FA tracks the active circulation in the inner retina using fluorescence. After being injected intravenously, sodium fluorescein reaches the ophthalmic artery, and the fundus camera excites it with blue light (peak absorption around 490 nm). The dye's electrons jump to a higher energy state, then relax and emit lower-energy yellow-green light at around 520 nm. A barrier filter blocks the original excitation wavelength so only the dye's emitted light reaches the sensor.

How useful FA is depends on the pharmacological properties of fluorescein itself. The molecule is small (376 Da) and binds reversibly to serum albumin, leaving a meaningful free fraction circulating in plasma. Normally, tight junctions in retinal capillaries keep both bound and free dye contained within the vessels. But in

conditions like DME or active CNV, where severe ischemia and elevated VEGF disrupt these junctions, fluorescein leaks into the surrounding tissue, showing up on FA as a slowly expanding bright cloud that confirms the inner BRB has broken down [16, 17].

Despite its sensitivity to leakage, FA has a major optical limitation. The blue-green wavelengths it uses overlap with the absorption spectrum of melanin and the macular xanthophylls, so the pigmented RPE absorbs much of the signal, hiding the choroidal circulation beneath it.

To get past this, Indocyanine Green Angiography (ICGA) was developed. ICGA uses a dye (775 Da) that absorbs and emits in the near-infrared range (peak absorption around 805 nm, peak emission around 835 nm). Near-infrared light passes through RPE melanin, xanthophyll, and even localized hemorrhage with little scattering [17]. Once injected, ICG binds strongly (about 98%) to plasma lipoproteins, mostly HDL, and this large complex can't pass through the 60 to 80 nm fenestrations of the choriocapillaris.

This combination of deep tissue penetration and staying within blood vessels lets ICGA reveal the choroidal vasculature clearly, without the background haze that obscures it on FA. ICGA remains the gold standard for visualizing deep, complex neovascular lesions, particularly the branching networks seen in inflammatory CNV and the dilated choroidal vessels typical of polypoidal choroidal vasculopathy [10, 17].

However, CFP, FA, and ICGA all share the same fundamental problem: they're two-dimensional. FA can confirm that leakage is happening, but it can't tell whether the abnormal membrane sits in the sub-RPE space or the subretinal space. Pinning down exactly where a lesion sits, how much fluid it's producing, and planning surgical or drug treatment all required moving beyond flat photography toward true depth-resolved imaging.

Relying entirely on flat, two-dimensional imaging created a real bottleneck in retinal care. While CFP and FA map a pathology's position in the x,y plane well, they mathematically squash all the depth information into a single value per pixel. This means planar imaging can't tell you where in the depth direction a lesion sits, so it's impossible to distinguish a shallow pocket of subretinal fluid from deep intraretinal cystic swelling or an organized sub-RPE membrane [6].

Solving this required a technology that could perform true optical sectioning without physically disturbing the tissue. The retina isn't flat, it's a three-dimensional structure, and disease often spreads vertically before it spreads sideways: inflammation, lipid leakage, and layer separation all expand depth-wise first. Capturing this

depth information is what Optical Coherence Tomography (OCT) was built to do, using localized, depth-resolved interferometry instead of wide-field projection [23].

OCT works by generating one-dimensional depth profiles called A-scans. As the beam scans across the retina, thousands of these A-scans combine into a detailed two-dimensional cross-section, or B-scan. The real breakthrough comes from the next step: scanning these B-scans across the y-axis builds a stack of cross-sections into a full three-dimensional data cube (C-scan) of the macula [23, 24].

This shift to volumetric imaging replaced rough visual estimates with precise measurements that directly inform treatment decisions. AMD assessment is a good example. Clinicians used to estimate severity by measuring how much area drusen covered on a fundus photo. But the metabolic and mechanical stress drusen place on the RPE depends on their actual volume, not just their footprint. Volumetric OCT lets algorithms trace the RPE and Bruch's membrane and calculate exact drusen height and volume across hundreds of B-scans, tracking changes down to the picoliter [11, 20].

Similarly, managing Diabetic Macular Edema (DME) depends heavily on depth-resolved data. A 2D angiogram can confirm that leakage is happening, but it gives no information about where that fluid actually ends up within the retina. Volumetric OCT resolves this directly, mapping the volume of intraretinal cysts (IRC) within the inner nuclear layer and separately identifying pockets of subretinal fluid (SRF) above the RPE [15]. Since IRC and SRF respond differently to steroids versus anti-VEGF treatment, measuring these volumes separately is essential for choosing the right treatment and predicting outcomes for the fovea [15, 16].

2.3.2 Optical Coherence Tomography Angiography (OCTA)

While volumetric structural OCT solved the depth problem for structural pathology, evaluating blood flow still relied on invasive dye-based imaging. Fluorescein Angiography (FA) and Indocyanine Green Angiography (ICGA) show vascular flow but have real optical drawbacks: they compress 3D capillary networks into a flat projection, and dye leakage tends to obscure the underlying vessels. Optical Coherence Tomography Angiography (OCTA) avoids these issues entirely by using the natural motion of red blood cells, producing depth-resolved, dye-free images of blood vessels through statistical variance analysis [5].

(a) The Physics of Motion Contrast and Decorrelation

OCTA relies on motion contrast. The system captures several closely spaced repeated B-scans (typically 2 to 4, called an M-B cluster) at the same x, y location over a very short time interval Δt [14]. Static tissue, like the structural matrix of the retina, scatters light consistently across these repeated scans, so the signal from

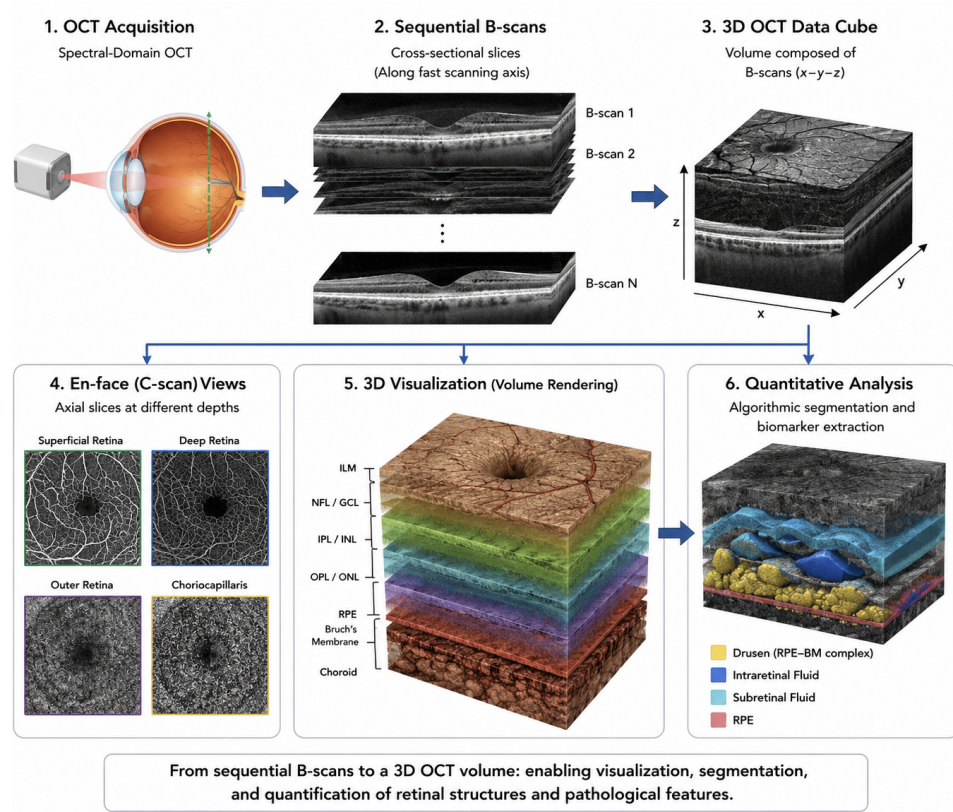


Figure 2.10: Conceptual workflow of OCT volumetric imaging

static regions stays nearly the same each time.

Red blood cells moving through capillaries, however, behave differently. As they move through the beam during Δt , they cause measurable fluctuations in both the amplitude and phase of the backscattered light [5]. The main computational task in OCTA is separating this dynamic signal from the static background.

The raw signal has both amplitude and phase information. Early OCTA approaches used phase-variance methods, relying on the Doppler phase shift $\Delta\phi$ caused by moving particles:

$$\Delta\phi = \frac{4\pi n v \cos(\theta) \Delta t}{\lambda_0}$$

where n is the tissue's refractive index, v is the red blood cell velocity, θ is the Doppler angle, and λ_0 is the laser's central wavelength. The problem with phase-based methods is that they're very sensitive to bulk tissue motion; even small patient movements create phase shifts that look like flow, degrading image quality.

To get around this, modern systems instead use amplitude-decorrelation methods, most notably Split-Spectrum Amplitude-Decorrelation Angiography (SSADA) [14]. SSADA ignores the fragile phase information entirely and looks only at how

intensity varies between scans. It splits the broad light source spectrum into narrower bands, which slightly reduces axial resolution for each band but creates multiple independent views of the same location. The algorithm calculates how different the sequential B-scans are for each spectral band, then averages these differences across all bands, which suppresses static noise without needing extra scan time. The resulting decorrelation signal correlates directly with moving red blood cells, and applying a threshold lets the software remove static tissue entirely, leaving a clear three-dimensional map of functional blood vessels.

(b) En-Face Projection and Capillary Plexus Mapping

The raw OCTA data is a set of isolated flow voxels. To make this useful clinically, software projects this flow data onto two-dimensional en-face images. This requires accurate structural segmentation, the algorithm finds reflective boundaries like the ILM, IPL, and INL, and defines depth "slabs" between them [13]. Within each slab, the strongest decorrelation signal along the z -axis is projected onto a flat image, separating the overlapping vascular layers into their own views [5].

The Radial Peripapillary Capillary (RPC) Plexus: The RPC plexus is a specialized capillary network found only in the thickest parts of the RNFL around the optic nerve head [5]. Unlike the winding, interconnected capillaries elsewhere in the retina, RPCs run in long, straight, parallel segments that follow the arc-shaped paths of the RGC axons. This layout meets the heavy metabolic needs of the unmyelinated axons as they converge toward the optic nerve. Because RPCs sit in the most superficial part of the retina, their density can be mapped very precisely with OCTA, making it a useful marker for ischemic axon loss in advanced glaucoma [5].

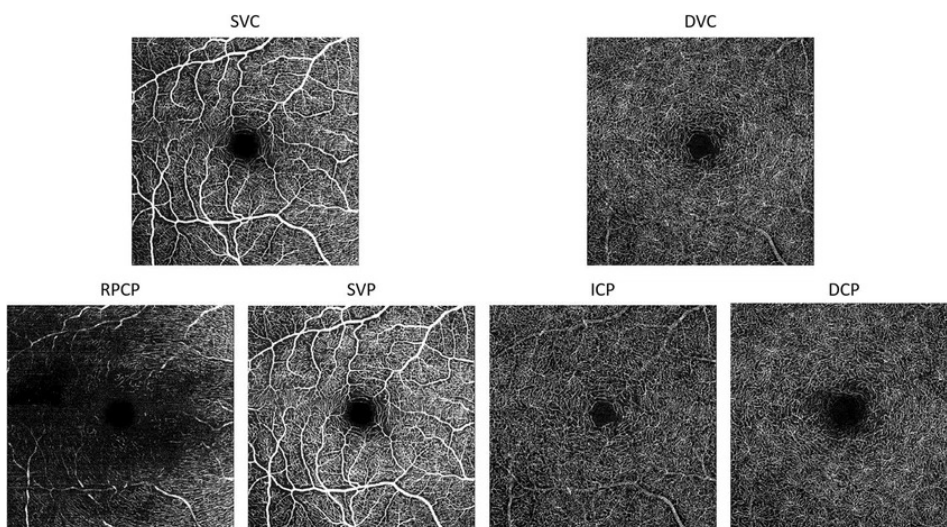


Figure 2.11: Topographical en-face OCTA mapping of retinal vascular plexuses

The Superficial Capillary Plexus (SCP): The SCP is the main network supplying arterial blood to the inner retina. Fed by precapillary arterioles from the central retinal artery, it sits mostly within the GCL and reaches inward to the front edge of the IPL [5, 13]. On OCTA, the SCP appears as a dense, flat, "spiderweb-like" pattern, with relatively large-diameter capillaries feeding the vertical connections that dive deeper into the retina.

The Intermediate and Deep Capillary Plexuses (ICP and DCP): Standard FA can't isolate deep retinal circulation, since any leakage from superficial vessels immediately obscures it. OCTA, however, can separate the Deep Vascular Complex (DVC) into two distinct networks bordering the INL. The Intermediate Capillary Plexus (ICP) sits at the inner edge of the INL, near the IPL, while the Deep Capillary Plexus (DCP) sits at the outer edge, near the OPL [5]. The DCP mostly acts as a drainage network, collecting deoxygenated blood from above and routing it to larger veins. Because it sits at the far end of the retina's oxygen supply chain and has no direct arterial feed, the DCP is especially vulnerable to hemodynamic failure. When this happens, the resulting ischemia can cause structural damage to the INL, a condition known as Paracentral Acute Middle Maculopathy (PAMM).

The Foveal Avascular Zone (FAZ) and Avascular Outer Retina: The central foveola has no inner retinal layers at all, in order to maximize light capture, leaving a capillary-free zone called the Foveal Avascular Zone (FAZ), usually 450 to 600 μm across [5]. On FA, the FAZ's edges are blurred by background choroidal fluorescence. OCTA removes this background completely, producing a sharp, high-contrast outline of the surrounding capillary ring. Precisely measuring the FAZ's area, perimeter, and roundness has become a standard way to track progressive microvascular loss in ischemic retinopathies [13].

OCTA also confirms an important physiological rule: the outer retina, comprising the ONL, ELM, and photoreceptor IS/OS layers, should be completely free of blood vessels. In a healthy eye, an OCTA slab from this region shows zero flow signal. Any detectable branching flow in this outer retinal slab is direct evidence of Choroidal Neovascularization (CNV) invading the area [14]. Being able to isolate this abnormal flow in three dimensions, without the confusion caused by pooled dye, changed how neovascular AMD is managed clinically.

2.4 Physics and Engineering of OCT

Optical Coherence Tomography combines the principles of optical interferometry with advanced signal processing to generate high-resolution cross-sectional im-

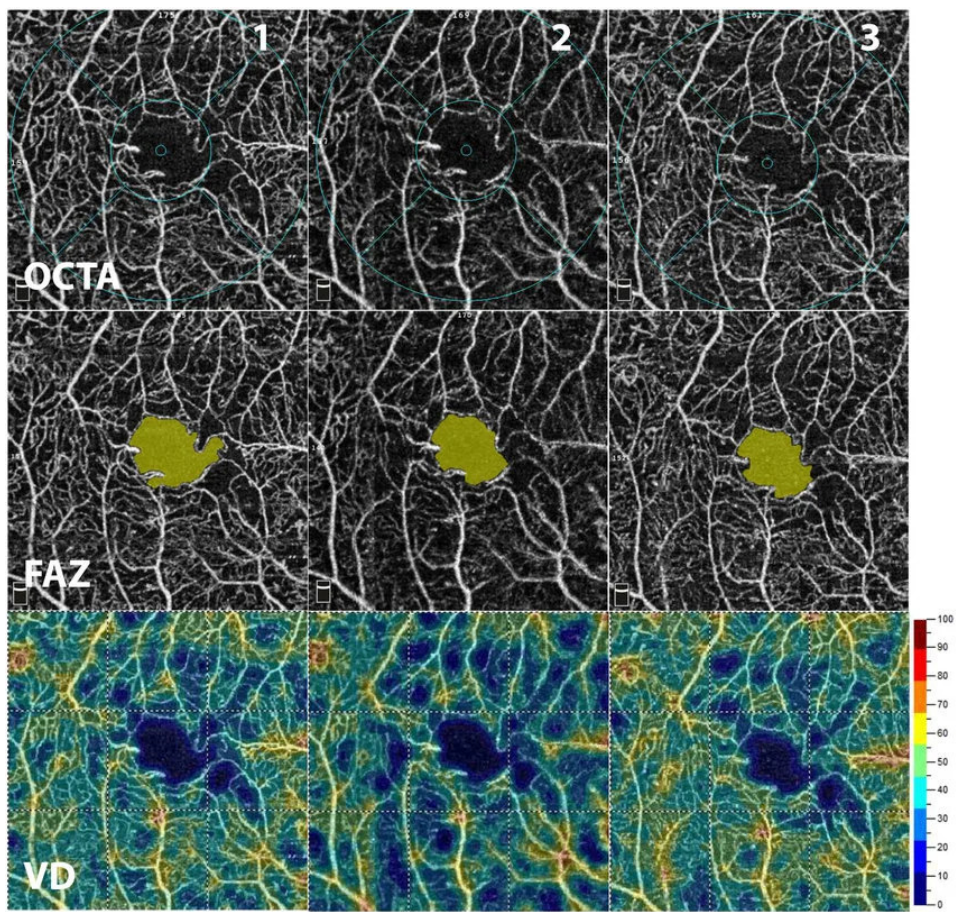


Figure 2.12: Topographical en-face OCTA mapping of the distinct retinal and choroidal vascular plexuses.

ages of biological tissue. Its ability to visualize retinal microstructure non-invasively has made it one of the most important imaging technologies in ophthalmology. Understanding its underlying physical principles provides the basis for interpreting OCT images and appreciating their diagnostic capabilities.

2.4.1 Low-Coherence Interferometry

Optical Coherence Tomography (OCT) is essentially the optical equivalent of ultrasound imaging. But there's a key difference that prevents directly copying ultrasound's approach: the speed at which the signal travels. Sound moves through tissue at around 1,500 m/s, slow enough for electronic systems to easily time returning echoes. Light, on the other hand, travels at 3×10^8 m/s in a vacuum and about 2.2×10^8 m/s through the eye's fluid. Measuring micrometer-scale structures by timing light directly would require detectors with femtosecond (10^{-15} s) resolution, far beyond what electronics can do. So OCT avoids direct timing altogether and instead measures the optical phase delay of reflected light using low-coherence interferometry [23, 24].

The core design of an OCT system is based on the Michelson interferometer. It starts with a broadband, low-coherence light source, historically a superluminescent diode (SLD). This light goes into a fiber-optic beam splitter (usually 50/50), which divides it into two paths: a reference arm and a sample arm.

The reference arm sends light to a mirror at a known, sometimes adjustable, distance. The sample arm sends light through the eye's optics and onto the layered retina. As the beam moves through the tissue, it scatters at every point where the refractive index changes, particularly at boundaries between cell layers like the ILM, RPE, and choriocapillaris.

Light scattered back from the retina and light reflected from the reference mirror travel back and recombine at the beam splitter. These combined fields, the reference field E_r and the sample field E_s , then hit a photodetector, which can only measure their time-averaged intensity I_d :

$$I_d = \langle |E_r + E_s|^2 \rangle = \langle E_r^2 \rangle + \langle E_s^2 \rangle + 2\langle E_r E_s \cos(2k\Delta z) \rangle \quad (2.1)$$

The first two terms, $\langle E_r^2 \rangle$ and $\langle E_s^2 \rangle$, are just constant background intensity from each arm and carry no depth information. All the useful depth information is in the third term, the interference, or cross-correlation, term.

Interference only happens when light from the sample and reference arms is mutually coherent, meaning their path lengths match closely enough. In a low-coherence system using a broadband SLD, the coherence length (l_c) is deliberately

very short, so the interference term only shows up when the reference arm's path length (z_r) and the sample arm's path length (z_s) match within that coherence length ($\Delta z = |z_r - z_s| \leq l_c$) [23].

This is what sets OCT's axial (depth) resolution. In normal microscopy, axial and lateral resolution are linked through the lens's numerical aperture, but OCT decouples them. Axial resolution (δz) in OCT depends only on the light source's spectral properties, following from the Wiener-Khinchin theorem. Assuming a Gaussian spectrum, axial resolution in tissue is:

$$\delta z = \frac{2 \ln 2}{\pi} \frac{\lambda_0^2}{n \Delta \lambda} \quad (2.2)$$

where λ_0 is the source's central wavelength, $\Delta \lambda$ is its bandwidth (FWHM), and n is the retina's average refractive index (about 1.36 to 1.38). This tells us how to improve depth resolution: it scales inversely with bandwidth [6, 24]. To get the roughly 5 μm axial resolution typical of modern clinical SD-OCT, manufacturers use broadband SLDs centered around 840 nm with bandwidths over 50 nm.

While axial resolution comes from spectral properties, lateral resolution (δx) is governed by ordinary diffraction and Gaussian beam optics, the smallest focal spot the eye's optics can produce on the retina:

$$\delta x = \frac{4 \lambda_0 f}{\pi d} \quad (2.3)$$

where f is the eye's focal length (about 16.7 mm) and d is the diameter of the light entering the pupil. To improve lateral resolution, you'd want to increase d , but this comes with a tradeoff related to the Rayleigh range (z_R), which sets the depth of focus:

$$b = 2z_R = \frac{\pi (\delta x)^2}{2 \lambda_0} \quad (2.4)$$

Depth of focus shrinks quadratically as lateral resolution improves. If engineers tried to push lateral resolution down to 1 μm , depth of focus would collapse to under 10 μm . Since the macula spans roughly 300 to 400 μm from the ILM to the choriocapillaris, such a shallow focus would leave most of the retina out of focus. For this reason, clinical OCT systems are intentionally

2.4.2 Time-Domain vs. Spectral-Domain OCT

The history of clinical OCT is largely a story of moving from mechanical to optical depth measurement, a shift that massively improved both speed and image quality. Early Time-Domain OCT (TD-OCT) systems used a classic Michelson in-

terferometer where the reference mirror was physically moved. To capture a single A-scan, the mirror had to mechanically travel a distance corresponding to the full depth being imaged.

This mechanical movement limited TD-OCT's speed severely. Because the mirror needed time to accelerate, move, and decelerate for every single A-scan, these systems topped out at around 400 to 800 A-scans per second. This was too slow to build dense volumetric scans; patients had to hold still for long periods while the system collected only a handful of sparse cross-sections. The slow speed also meant the system integrated the signal over a long time window, making TD-OCT images very prone to motion artifacts, even small tremors or the pulse from blood flow could blur and disrupt a B-scan [6, 24].

Spectral-Domain OCT (SD-OCT) got rid of the moving mirror entirely, replacing it with a fixed reference arm and a spectrometer-based detector. The light source stays broadband, but the returning interference signal is sent through a diffraction grating, which spreads it into its component wavelengths and focuses them onto a fast, multi-pixel linear CMOS array.

By capturing the whole interference spectrum at once, SD-OCT encodes depth information as the frequency of the fringe pattern. A backscattering boundary at depth Δz produces a modulation frequency proportional to Δz . The system then extracts the depth profile by applying an inverse Fast Fourier Transform (iFFT) to the captured spectrum [24].

This change brought two major benefits. First, it eliminated the mechanical speed limit entirely, pushing acquisition speeds to 50,000 to 100,000 A-scans per second. This made it possible to capture dense 3D volumes in a fraction of a second, effectively freezing eye motion and enabling the automated segmentation that modern AI diagnostics rely on [23]. Second, SD-OCT gained a built-in sensitivity advantage: because it detects all spectral components at once rather than sequentially, it doesn't need to integrate the signal over the full depth range, which dramatically improves the signal-to-noise ratio. This lets SD-OCT pick up subtle, low-reflectivity boundaries, such as the outer edge of the ellipsoid zone, that were essentially invisible to TD-OCT.

2.5 Artificial Intelligence in Ophthalmology

Artificial Intelligence has become an increasingly important component of ophthalmic research and clinical practice by enabling automated analysis of retinal imaging data. The availability of large annotated datasets and advances in computational hardware have accelerated the development of intelligent diagnostic systems capable of assisting clinicians in disease detection, classification, and decision

support.

2.5.1 The Clinical Data Bottleneck

Modern ophthalmology has a mismatch between how much data it can collect and how much a human can realistically review. A single scan from a high-speed Swept-Source OCT device produces a cube of 500 to 1000 B-scans, each containing 1024 A-scan depth profiles, more than 500,000 individual depth measurements per eye, per visit. Across a clinic managing thousands of patients with chronic conditions like AMD or diabetic retinopathy, this volume of structural and angiographic data quickly exceeds what any clinician can manually review.

The bottleneck isn't collecting the data, it's making sense of it. Clinicians currently rely on simplified metrics like Central Subfield Thickness (CST) as a stand-in for disease status. But CST compresses a lot of information into one number and hides important structural details. It can't tell the difference between a harmless variation in foveal shape and a clinically significant fluid buildup that needs urgent anti-VEGF treatment. Relying on such compressed numbers means ignoring the rich spatial information in the OCT volume, leading to delayed diagnoses, inconsistent staging, and treatment that isn't well-tuned to the individual.

This is essentially why AI is needed in this field. Deep learning offers a practical way to unpack these high-dimensional datasets, turning raw interference signals into measurable, clinically meaningful structural markers that can be tracked over time [21, 24].

2.5.2 Machine Learning vs. Deep Learning

Moving from traditional machine learning to deep learning represents a fundamental change in how features are extracted. Early approaches to OCT analysis relied on manually designed features, things like Gray-Level Co-occurrence Matrices, Gabor filters, or wavelet transforms, to detect patterns such as drusen or cyst boundaries. This required experts to manually choose which filters or texture descriptors they thought captured the relevant pathology. These methods tend to be fragile, easily thrown off by image noise, speckle, and differences in image contrast between manufacturers like Zeiss and Heidelberg.

Deep learning, particularly Convolutional Neural Networks (CNNs), removes this manual step by learning useful representations automatically. A CNN isn't given a pre-built "drusen filter"; instead, it's trained directly on raw OCT B-scan pixels. Through backpropagation across millions of parameters, the network learns its own hierarchy of filters suited to the task. Early layers tend to pick up simple features like edges and intensity boundaries, such as the bright RPE line, while

deeper layers combine these into more complex patterns, like the shape of a CNV complex or the volume of intraretinal cystic fluid [14].

2.6 Challenges in Explainable AI for OCT Classification

Optical Coherence Tomography (OCT) has become the gold standard imaging modality for the diagnosis and monitoring of retinal diseases because of its ability to produce high-resolution cross-sectional images of retinal microstructures. The detailed visualization of retinal layers enables ophthalmologists to detect pathological changes associated with diseases such as Age-Related Macular Degeneration (AMD), Diabetic Macular Edema (DME), and Choroidal Neovascularization (CNV). Despite its diagnostic capabilities, the interpretation of OCT images presents several challenges that have motivated the adoption of Artificial Intelligence (AI) and, more recently, Explainable Artificial Intelligence (XAI).

Traditionally, OCT images are interpreted manually by trained ophthalmologists. This process requires careful examination of multiple retinal layers to identify subtle structural abnormalities such as intraretinal fluid, subretinal fluid, pigment epithelial detachments, or drusen deposits. Since these pathological changes may be small and exhibit similar visual characteristics across different retinal diseases, accurate diagnosis largely depends on the clinical expertise and experience of the physician. Furthermore, the increasing number of OCT examinations performed in modern ophthalmic practice has substantially increased the workload on clinicians. Manual assessment of large volumes of OCT scans is therefore time-consuming and may lead to inter-observer variability, where different specialists may provide slightly different interpretations of the same image based on their individual clinical experience.

To overcome these limitations, Artificial Intelligence, particularly deep learning, has been widely adopted for automated OCT image classification. AI-based systems can rapidly analyze large numbers of retinal images and assist clinicians by providing highly accurate disease predictions. These systems have demonstrated significant potential for improving screening efficiency, reducing diagnostic workload, and supporting early detection of retinal diseases. However, despite their excellent predictive performance, many AI models generate only a diagnostic label without explaining how the decision was reached. In clinical practice, a prediction alone is often insufficient, as ophthalmologists require evidence that the diagnosis corresponds to observable retinal abnormalities before making treatment decisions.

This lack of transparency represents one of the primary challenges in the clinical adoption of AI-assisted diagnostic systems. Medical professionals are generally more likely to trust an automated prediction when they can understand the reason-

ing behind it and verify that it is consistent with established clinical knowledge. In ophthalmology, where treatment decisions directly influence patient outcomes, clinicians must be able to determine whether an AI model has focused on the relevant retinal structures and pathological features rather than unrelated image characteristics. Consequently, transparency and interpretability have become essential requirements for the practical deployment of AI systems in healthcare.

Explainable Artificial Intelligence (XAI) has emerged as an important approach for addressing these challenges by improving the interpretability of AI-based diagnostic systems. Rather than replacing clinical expertise, XAI aims to complement the diagnostic process by providing explanations that help clinicians understand and validate model predictions. Such explanations can improve confidence in automated systems, facilitate collaboration between clinicians and AI models, and promote safer integration of artificial intelligence into routine ophthalmic practice. As the use of AI in retinal disease diagnosis continues to expand, explainability is expected to play an increasingly important role in developing trustworthy and clinically acceptable diagnostic solutions.

LITERATURE REVIEW

2.7 Literature Review

The rapid growth of Explainable Artificial Intelligence has resulted in a substantial body of research on improving the transparency of deep learning models for retinal disease classification. Examining the existing literature provides the necessary context for understanding current advances, identifying recurring limitations, and motivating the proposed work.

2.7.1 Research Problem Formulation

The deployment of deep learning architectures for the diagnosis of retinal pathologies using Optical Coherence Tomography (OCT) is fundamentally constrained by the "black-box" nature of artificial neural networks. While modern convolutional and transformer-based models achieve classification accuracies that rival or exceed those of trained ophthalmologists, they obscure the specific morphological biomarkers driving their predictions. This lack of transparency conflicts with the medicolegal requirements of clinical ophthalmology, where treatment protocols such as intravitreal anti-VEGF injections demand explicit, anatomically verifiable justification rather than an unexplained diagnostic label.

Explainable AI has been proposed as the mechanism through which this justification can be recovered, typically by generating a spatial relevance map that is superimposed over the original B-scan to indicate which regions of the image most strongly influenced the network's decision. Gradient-weighted Class Activation Mapping remains the most widely adopted technique for this purpose across the reviewed literature, owing to its compatibility with standard convolutional architectures and its comparatively low implementation overhead. However, as established in the preceding sections, the manner in which such explanations are conventionally derived, predominantly from the terminal convolutional layer of the network, introduces systematic limitations in spatial resolution, boundary precision, and robustness to input perturbation.

The research problem addressed by this study is therefore twofold. First, there is a need to systematically characterize the extent to which existing deep learning classification architectures for OCT-based retinal disease diagnosis have been coupled with, and evaluated alongside, explainability mechanisms, since a substantial portion of the literature continues to report classification performance in isolation from any interpretability analysis. Second, among the studies that do incorporate an ex-

plainability component, there is a need to critically examine how consistently these components are evaluated, given the documented lack of a coherent, standardized framework for quantifying explanation quality. Establishing a clear, evidence-based picture of both dimensions, classification performance and explainability rigor, is a necessary precondition for identifying the specific methodological gaps that current approaches leave unresolved.

Formally, Let the classifier be represented by $f: \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X}$ denotes the OCT image space and $\mathcal{Y}$ the disease label space, while the corresponding explanation generator is $E: \mathcal{X} \rightarrow \mathcal{H}$, where $\mathcal{H}$ represents the generated heatmap. The objective is to maximize both the classification performance, $P(f)$, and the explanation quality, $Q(E)$, such that

$$\max \mathcal{J} = \alpha P(f) + \beta Q(E), \quad \alpha, \beta \geq 0, \alpha + \beta = 1,$$

where existing approaches predominantly optimize $P(f)$ while providing limited or non-standardized evaluation of $Q(E)$. Consequently, the research seeks to identify and analyze this imbalance through a systematic review of OCT-based deep learning literature.

2.7.2 The Paradigm of Explainable AI (XAI) in Ophthalmology

To bridge the gap between mathematical optimization and clinical trust, researchers introduced spatial attribution methods designed to visually interpret the network's latent feature space. The most prominent technique, Gradient-weighted Class Activation Mapping (Grad-CAM), operates by computing the gradients of the target concept (the disease label) flowing into the final convolutional layer of the network. These gradients are globally average-pooled to obtain neuron importance weights, which are subsequently multiplied by the forward activation maps to generate a coarse, localized heatmap highlighting the highly discriminative regions.

The application of Grad-CAM in ophthalmic imaging rapidly expanded. Recent frameworks successfully coupled multi-branch CNNs with Grad-CAM visualization to isolate distinct morphological asymmetries, allowing clinicians to verify whether the network was correctly isolating subretinal fluid or erroneously focusing on imaging artifacts [25, 26]. Explainable networks were also deployed to investigate acute-stage fluorescein angiography and widefield structural OCT, successfully predicting long-term visual prognoses by highlighting regions of severe macular non-perfusion and neurodegeneration [2, 27]. In the domain of fundus analysis, the integration of the Convolutional Block Attention Module (CBAM) with CheXNet models significantly refined the spatial boundaries of the Grad-CAM heatmaps,

forcing the network to highlight precise microaneurysms and hard exudates associated with diabetic retinopathy [28].

Despite the widespread clinical adoption of these visual explanations, a severe foundational flaw remains regarding their quantitative reliability. A critical investigation assessing the coherency of visual explanation metrics revealed that standard evaluation methodologies often yield contradictory rankings. Metrics designed to quantify the "goodness" of a heatmap, such as Insertion and Deletion scores, encode fundamentally opposing notions of relevance, while proxy-task evaluations (e.g., bounding box intersection) exhibit negligible correlation with the actual internal logic of the model [3].

Furthermore, the direct applicability of these natural-image attribution methods to the medical domain is highly contested. An applicability study evaluating DeepLIFT, AttEXplore, and Grad-CAM on widefield OCTA and structural OCT images demonstrated that while the baseline VGG16 model achieved extreme accuracy ($> 98\%$) for fluid detection, the resultant attribution heatmaps yielded wildly divergent spatial localizations [29]. Because the spatial coarseness and mathematical sparsity of the generated explanations dictate the evaluation scores, relying solely on uncalibrated Grad-CAM projections poses a severe medicolegal risk. The visual output may highlight a clinically relevant biomarker, but it cannot guarantee that the specific biomarker was the actual mathematical determinant for the network's prediction.

2.7.3 Systematic Literature Review (SLR) Methodology

To rigorously assess the current landscape of pixel-level interpretability in OCT-based retinal disease classification and address the quantitative inconsistencies identified in the literature, we executed a Systematic Literature Review (SLR) adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

The primary objective of this SLR was to isolate, extract, and synthesize peer-reviewed publications that deployed deep learning models for multi-class retinal disease detection (focusing on AMD, DME, CNV, and normal controls) while explicitly implementing XAI visualization techniques, predominantly Grad-CAM. The search protocol interrogated major academic databases, including IEEE Xplore, ScienceDirect, SpringerLink, and Nature, restricting the temporal inclusion boundary to the last five years to capture the modern architectural transition from standard CNNs to attention-guided transformers.

Search queries utilized complex Boolean strings crossing technical domains, organized by category and target database as summarized in Table 2.1.

Table 2.1: Search Queries Used in the Systematic Literature Review

Category	Search Query
Retinal Disease Papers	(retinal disease OR macular degeneration OR diabetic macular edema) AND optical coherence tomography
OCT Classification Papers	(OCT OR optical coherence tomography) AND (classification OR deep learning)
CNN Papers	(convolutional neural network OR CNN) AND retinal AND OCT
Explainable AI Papers	(explainable AI OR XAI OR interpretability) AND retinal disease
Grad-CAM Papers	(Grad-CAM OR class activation mapping) AND (retina OR OCT OR fundus)
XAI Evaluation Papers	(evaluation OR coherency OR faithfulness) AND (explainable AI OR saliency OR attribution)

Studies were subjected to strict inclusion criteria: 1. The research must present an automated diagnostic pipeline utilizing human OCT or fundus imagery. 2. The model must incorporate an explicit Explainable AI or attribution-based interpretability component. 3. The methodology must feature empirical evaluation of the network architecture and the resulting explanation.

To reflect the professor’s guidance that the review should focus exclusively on Explainable AI contributions rather than general classification architectures, the extracted corpus was restricted to studies that explicitly implement or evaluate an interpretability technique. This corpus was then stratified by the primary attribution mechanism employed: studies centered on Class Activation Mapping and its direct variants (Table 2.3), and studies employing alternative, hybrid, or model-agnostic explainability approaches, including perturbation-based, gradient-based, and quantitative evaluation methods (Table 2.4).

2.7.4 Inclusion and Exclusion Criteria

To ensure that the corpus synthesized in this chapter reflects work that is both methodologically sound and directly relevant to pixel-level interpretability in retinal disease classification, every candidate study identified through the search strategy in Table 2.1 was screened against a structured set of inclusion and exclusion criteria. Table 2.2 presents these criteria in full.

Table 2.2: Inclusion and Exclusion Criteria Applied to Candidate Studies

Code	Criterion
Inclusion Criteria	
IC1	The study presents an automated diagnostic pipeline applied to human OCT imagery, or to closely related ophthalmic imagery (fundus photography, OCTA, fluorescein angiography) directly relevant to retinal disease classification.
IC2	The study incorporates an explicit Explainable AI or attribution-based interpretability component (Class Activation Mapping and its variants, perturbation-based methods, gradient-propagation methods, or hybrid fusion techniques), rather than reporting classification performance in isolation.
IC3	The study reports empirical evaluation, whether quantitative or qualitative, of the classification architecture and of the resulting explanation, permitting extraction of a discrete finding regarding the explainability technique itself.
IC4	The study is available as a peer-reviewed publication or a technically complete preprint published within a recency window judged sufficient to reflect the current architectural and methodological state of the art.
Exclusion Criteria	
EC1	The study addresses retinal disease classification using deep learning but provides no explainability or interpretability component whatsoever. Studies meeting this exclusion, such as [30], [31], [32], [33], and [34], report competitive classification performance but were excluded because they offer no attribution, saliency, or explanation output to synthesize against the review’s XAI-centered scope.
EC2	The study applies an explainability technique exclusively to an imaging domain unrelated to ophthalmology, and does not contribute a methodologically foundational evaluation framework of direct relevance to medical image interpretability. Studies failing this criterion were excluded unless, as with [3], the study’s evaluation-coherency findings apply directly to the metrics used elsewhere in the retina-specific literature.
EC3	The study duplicates the dataset, backbone architecture, and explainability technique of an already-included study without introducing a distinguishable methodological or evaluative contribution.
EC4	The study does not disclose sufficient methodological detail, such as an abstract-only record or a non-technical short communication, to determine with confidence which specific explainability technique was employed.

Applying EC1 in particular removed a substantial share of otherwise well-performing classification studies from the reviewed corpus, since a considerable portion of the retinal OCT deep learning literature continues to report multi-class diagnostic performance without any accompanying interpretability analysis. The remaining criteria, applied jointly, ensured that every study retained for synthesis in Table 2.3 and Table 2.4 contributes a discrete, extractable finding about how explainability was implemented, evaluated, or critiqued, rather than treating XAI as an incidental figure attached to an otherwise unrelated classification contribution.

2.7.5 Research Questions

Building upon the research problem and objectives established in 1, the systematic literature review is guided by the following research questions. These questions provide a structured framework for investigating the current state of research in OCT-based retinal disease classification and Explainable Artificial Intelligence. The findings corresponding to these questions are discussed in 2.8.

1. **RQ1:** *What publicly available and clinically acquired datasets have been used for the development, training, validation, and evaluation of deep learning models for OCT-based retinal disease classification, and what are their key characteristics, advantages, and limitations in supporting reliable and generalizable diagnostic systems?*
2. **RQ2:** *What deep learning architectures and classification techniques have been proposed for OCT-based retinal disease classification, and how have these approaches evolved to improve diagnostic accuracy, computational efficiency, and clinical applicability?*
3. **RQ3:** *What Explainable Artificial Intelligence (XAI) techniques and evaluation methodologies have been employed to interpret and validate deep learning models for OCT-based retinal disease classification, and what challenges remain in establishing reliable, robust, and clinically trustworthy explanations?*

2.7.6 Literature Synthesis and Analysis Tables

Table 2.3 synthesizes the studies that rely on Class Activation Mapping and its direct architectural variants (Grad-CAM, Grad-CAM++, Score-CAM) as the primary explainability mechanism for retinal disease classification. Table 2.4 synthesizes the remaining studies, which employ alternative, hybrid, or model-agnostic explainability approaches, including perturbation-based attribution, relevance propagation, and quantitative evaluation frameworks that assess CAM-family methods rather than deploying them directly.

Table 2.3: Synthesis of Class Activation Mapping-Based Explainability Techniques

References	Focus Area	Key Results	Limitations
[35]	Emergency-setting detection of central retinal artery occlusion from OCT, with Grad-CAM used to inspect the discriminative regions of a ResNet-18 classifier.	Grad-CAM feature maps were consistent with clinically expected regions of ischemic change across a multi-center retrospective cohort.	Evaluated on a small dataset (293 images) with a limited range of differential diagnoses and high-quality imaging only, restricting generalizability.
[25]	Multiclass retinal disease classification from OCT using a 2D CNN, exploiting retinal asymmetry patterns and Grad-CAM for post-hoc visual explanation.	The classifier achieved very high benchmark accuracy, with Grad-CAM heatmaps qualitatively aligned with expected lesion regions.	Grad-CAM output was assessed only qualitatively rather than against clinical annotation, and evaluation was limited to public benchmark data.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[36]	Two-stage glaucoma identification pipeline combining Attention U-Net segmentation with InceptionV3 classification, explained using both Grad-CAM and Grad-CAM++.	Comparative use of two CAM variants provided complementary explanatory views of the same classification decision.	Annotated glaucoma datasets remain limited in size and diversity, and the black-box nature of the classifier is only partially addressed by CAM-based explanation.
[37]	Class-Re-Attention Transformer architecture for five-class OCT classification, with Grad-CAM used to visualize lesion-relevant regions.	The re-attention mechanism reduced the influence of speckle noise on the reported explanation while maintaining high classification accuracy.	Grad-CAM was applied qualitatively without quantitative validation, and the external test set was comparatively small.
[38]	Comparative evaluation of four pre-trained CNN backbones for OCT-based eye disease classification, incorporating latent-space class-imbalance correction and Grad-CAM visualization.	Latent feature-space oversampling improved minority-class performance while Grad-CAM confirmed attention to disease-relevant regions.	The dataset originated from a single clinical center and lacked demographic metadata, limiting assessment of potential bias.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[26]	Ocular tumor lesion localization and classification using Grad-CAM integrated directly into the training pipeline via a heatmap-aligned loss term.	Embedding Grad-CAM within training, rather than applying it purely post-hoc, improved the spatial correspondence between the heatmap and the lesion.	No expert-annotated pixel-level ground truth was available to quantitatively validate the improved localization.
[28]	Diabetic retinopathy classification using a CheXNet backbone augmented with the Convolutional Block Attention Module, explained via Grad-CAM.	Joint channel and spatial attention narrowed the Grad-CAM heatmaps toward clinically relevant microaneurysms and exudates.	Quantitative, annotation-based validation of the resulting heatmaps was not available for the public datasets used.
[1]	Joint retinal disease classification and localization using a novel CNN (FCNNplus), explained through Grad-CAM and Score-CAM with a proposed inter-method similarity index.	The similarity index provided a quantitative means of comparing agreement between two CAM variants applied to the same model.	Overall classification accuracy trailed some state-of-the-art transfer-learning models, and the similarity index itself was not externally validated.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[2]	Grad-CAM-based analysis of acute-phase fluorescein angiography to predict long-term visual prognosis following branch retinal vein occlusion.	Grad-CAM identified perifoveal vasculature as the most influential region for the prognosis prediction, offering a candidate anatomical biomarker.	The dataset was small and single-center, correlation between predicted and observed outcomes was only moderate, and the imaging modality was angiography rather than OCT.
[39]	Design of an ultra-lightweight CNN architecture for retinal disease classification suitable for edge deployment, with Grad-CAM applied for interpretability.	Substantial reductions in parameter count were achieved with limited loss of multi-class diagnostic accuracy.	Grad-CAM was applied without quantitative evaluation of heatmap quality, and performance degraded on more heterogeneous test sets.
[40]	Multi-CNN ensemble (Xception, VGG16, InceptionV3, DenseNet121) for diabetic retinopathy detection, with Grad-CAM used to support model interpretability.	Stacking generalization with a k-nearest-neighbor meta-learner improved classification accuracy while Grad-CAM provided qualitatively meaningful visual support.	The dataset was small, classification was restricted to a binary DR stage distinction, and Grad-CAM was evaluated only qualitatively.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[41]	General-purpose worked case study of the Grad-CAM mechanism applied to medical image classification.	The study demonstrates that Grad-CAM heatmaps can align with clinically meaningful regions, supporting the broader case for CAM-based trust-building.	The case study is not retina- or OCT-specific, evaluates a single technique qualitatively, and does not report a quantitative faithfulness metric.
[42]	OCT-based multi-disease classification combining squeeze-and-excitation attention with a dual explainability approach of Grad-CAM and gradient-based saliency maps.	The dual-explanation approach was demonstrated in a deployed prototype intended for practical clinical usability.	The two explanation methods were combined without a formal agreement or consistency metric between them.
[43]	Federated Faster R-CNN framework combining contour detection with Grad-CAM for interpretable multi-class retinal disease diagnosis under privacy-preserving distributed training.	The framework demonstrated that interpretability could be preserved alongside a federated, privacy-preserving training regime.	Only two federated clients were used, limiting assessment of scalability, and the evaluation was restricted to fundus imagery.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[44]	Proposes a quantitative, domain-grounded Grad-CAM evaluation metric for diabetic retinopathy detection, moving beyond purely qualitative heatmap inspection.	The proposed conformity metric quantified overlap between Grad-CAM activation and ground-truth lesion annotation, offering a reusable benchmarking tool.	The metric was validated on fundus imagery rather than OCT and requires costly expert-annotated lesion-level ground truth.
[45]	Ensemble transfer-learning model for ocular disease classification, with Grad-CAM applied to the ensemble output to validate clinical alignment of attention.	Grad-CAM-derived attention regions were reported to align closely with clinically relevant disease markers across the ensemble.	The work remains an unreviewed preprint restricted to fundus imagery, and applying Grad-CAM to an ensemble output raises open questions about attribution of the underlying gradients.
[46]	Ocular disease diagnosis centered on Grad-CAM as the sole explainability mechanism, presented with a detailed mathematical treatment of the technique.	The study offers one of the more rigorous methodological walkthroughs of the Grad-CAM formulation among the reviewed literature, validated on clinical imagery.	No comparison against other CAM variants was performed, and the imagery was general ocular rather than OCT-specific.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[47]	Hybrid CNN-Transformer network for OCT-based retinal disease classification, with Grad-CAM applied consistently across three public OCT benchmark datasets.	Grad-CAM validation across multiple OCT datasets directly supports the relevance of the architecture's attention to pixel-level OCT interpretability.	Evaluation was restricted to public datasets without clinical data, and only a dual-branch CNN-Transformer configuration was explored.
[48]	Multistage OCT disease screening, classification, and pathologic region detection pipeline, contrasting Grad-CAM heatmaps against explicit object-detection-based localization.	The comparison suggested that bounding-box-based localization can provide more clinically actionable spatial evidence than Grad-CAM heatmaps alone in some contexts.	Data originated from a single hospital, and bounding-box annotation is inherently less precise than pixel-level retinal-layer segmentation.
[49]	Lightweight, resource-constrained fundus-image screening model using both Grad-CAM and Grad-CAM++ jointly for point-of-care interpretability.	Dual CAM-variant usage ensured pixel-level visual evidence accompanied every classification decision in a compact architecture.	Evaluation relied on a single national, single-modality dataset without external multi-ethnic validation.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[50]	Comparative classification study across four CNN backbones on a regional (Ghanaian) OCT clinical dataset, interpreted using Grad-CAM and saliency maps.	Cross-architecture comparison broadened geographic representation in the OCT explainability literature while confirming consistent Grad-CAM behavior across backbones.	Accuracy gains and CAM interpretability were not evaluated with a joint quantitative agreement metric.
[51]	Diabetic retinopathy detection from angiography imagery with a quantitative comparison of Grad-CAM and Score-CAM using a purpose-built similarity index.	The proposed similarity index enabled direct, quantitative comparison of inter-CAM agreement on identical underlying models.	The similarity index measures inter-CAM agreement rather than agreement with expert-annotated ground truth, and the imaging modality was angiography rather than OCT.
[52]	Computer-aided diagnosis tool for OCT-based retinal disorder detection combining semantic segmentation-guided interpretability with Grad-CAM++.	Grad-CAM++ was shown to produce clearer heatmaps and regions of interest than standard Grad-CAM in this application.	Evaluation was limited to a subset of a single public dataset (UCSD) and only four disease classes.

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Table 2.3 continued from previous page

References	Focus Area	Key Results	Limitations
[53]	Ensemble eye disease detection combining fundus and vascular-structure inputs, using Grad-CAM comparatively across single- and dual-branch model configurations.	Grad-CAM comparison provided direct visual evidence that incorporating vascular-structure input measurably improved interpretability over single-input models.	The comparison was presented qualitatively via heatmap figures rather than through a formal quantitative metric, and only fundus imagery was used.
[54]	Regional (Ethiopian) fundus-image benchmark for early retinal disease detection, with Grad-CAM applied to the best-performing DenseNet201 model.	Grad-CAM validated that the best-performing model attended to disease-relevant regions, broadening geographic diversity in the Grad-CAM literature.	The dataset was single-institution and regionally specific, and only four disease classes were covered.

Table 2.4: Synthesis of Alternative Explainable AI Approaches in Retinal Disease Classification

References	Focus Area	Key Results	Limitations
[55]	Scoping review of Class Activation Mapping variants (Grad-CAM, Grad-CAM++, Score-CAM, Ablation-CAM, LayerCAM, and related methods) applied to diabetic retinopathy on fundus images.	Synthesized findings across 28 studies and 14 datasets, identifying recurring failure modes such as vessel-focused heatmaps, optic disc glare, and poor sensitivity to low-contrast microaneurysms.	Being a review rather than an original study, it reports no new experimental validation; it further notes that causal faithfulness evaluation of CAM methods remains comparatively underdeveloped.
[56]	OCT-based retinal disease classification explained using a perturbation-based, model-agnostic technique (LIME) rather than gradient-based CAM.	Demonstrated that superpixel-based perturbation explanations can be used to interpret a hybrid CNN-SVM classifier as an alternative to Grad-CAM.	Because the explanation is perturbation-based rather than gradient-weighted, direct comparability with the predominantly Grad-CAM-centric literature is limited.
[29]	Systematic comparison of DeepLIFT, AGI, and AttEXplore attribution methods within a unified VGG16 pipeline for diabetic retinopathy and fluid detection on OCT and OCTA imagery.	High classification accuracy did not translate into consistent attribution; the compared methods produced markedly divergent spatial localizations despite operating on the same underlying model.	Attribution maps did not constitute pixel-accurate segmentations, and variability across methods was high even under matched experimental conditions.

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Table 2.4 continued from previous page

References	Focus Area	Key Results	Limitations
[27]	Explainable AI-assisted exploration of diabetic retinal neurodegeneration on OCT, using relevance propagation (RelProp/LRP) rather than CAM-based attribution.	Achieved high discriminative performance for visual-acuity thresholding while relevance maps highlighted specific neuroretinal layers associated with vision loss.	The study design was single-center, evaluated only horizontal OCT sections from a specific device, and prior vitrectomy was a potential confounding factor.
[57]	Dual-backbone CNN fusion framework for multi-class eye disease classification, explained using Score-CAM together with LIME-based superpixel segmentation.	Systematic evaluation of twelve fusion configurations, combined with external validation, confirmed cross-dataset generalization alongside consistent Score-CAM and LIME explanations.	Fusion performance was architecture-dependent, and external datasets introduced domain-shift effects not fully resolved by the reported fusion strategies.
[3]	Large-scale comparison of visual explanation methods (Grad-CAM, Integrated Gradients, LRP, occlusion, RISE, SmoothGrad, TAME, adaSISE) and their quantitative evaluation metrics.	Demonstrated that standard evaluation metrics such as Insertion and Deletion scores frequently produce contradictory rankings of the same set of explanations.	Experiments were restricted to a subset of ImageNet validation images and two architectures, limiting direct transfer of conclusions to the medical imaging domain.

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Table 2.4 continued from previous page

References	Focus Area	Key Results	Limitations
[58]	Diabetic retinopathy grading from fundus images using a dual explainability approach combining Grad-CAM with the game-theoretic SHAP attribution method.	The combination of pixel-level CAM evidence and feature-level SHAP attribution provided complementary local and global explanations within an ensemble weight-fusion strategy.	The modality evaluated was fundus imagery rather than OCT, and the two explanation outputs were not accompanied by a formal quantitative agreement measure.
[59]	Glaucoma classification from fundus images explained using SCIM, a novel fusion of SHAP attribution values with CAM-based spatial localization, benchmarked against standard Grad-CAM.	The proposed hybrid technique was directly compared against Grad-CAM and SHAP individually, providing a comparative basis for evaluating fused explanation strategies.	Evaluation was restricted to fundus imagery, and no large-scale quantitative validation of the proposed fusion technique was reported.
[60]	Diabetic retinopathy detection combining Grad-CAM for spatial evidence with SHAP for feature-level quantification within a single diagnostic pipeline.	Demonstrated that combining spatial and feature-level explanation techniques improved the interpretability of the detection pipeline beyond either method used alone.	No formal quantitative agreement score was reported between the Grad-CAM and SHAP outputs, and the OCT modality was not addressed.

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Table 2.4 continued from previous page

References	Focus Area	Key Results	Limitations
[61]	Structured survey of Explainable AI techniques, including CAM-family methods, LIME, SHAP, and attention-based approaches, applied to diabetic retinopathy and related ophthalmic diseases.	Provided a taxonomy of XAI techniques together with a survey of publicly available fundus datasets suitable for reproducible explainability research.	As a review-level analysis, it reports no new experimental validation, and OCT-specific interpretability is only partially covered relative to fundus imaging.

2.8 Discussions

The preceding review assembled and synthesized a broad corpus of studies concerned with Explainable AI in OCT and fundus-based retinal disease classification, stratified in Table 2.3 and Table 2.4 by primary explainability mechanism. This section draws that material together to directly address the three research questions posed in Section 2.7.5.

RQ1: *What Types of Datasets Are Used in OCT Research?*

The reviewed corpus draws on a comparatively small set of recurring public benchmarks alongside a longer tail of institution-specific clinical datasets. Among public benchmarks, the Kermany/UCSD OCT dataset and its four-class (CNV, DME, Drusen, Normal) labeling scheme recurs across a large share of the reviewed studies, including [1] and [56], while the Duke OCT dataset appears as a secondary or cross-validation benchmark in [56]. More recent multi-dataset benchmarks, such as the OCT-2014, OCT-2017, and OCT-C8 collections used jointly by [47], extend coverage to eight-category disease taxonomies. Beyond conventional B-scan OCT, several studies draw on adjacent modalities: [29] incorporates widefield OCT Angiography (OCTA) imagery alongside standard OCT, [36] and [59] rely on fundus photography for glaucoma-related tasks, and [58] draws on multiple independent fundus datasets (APTOS, MESSIDOR, IDRiD) to test cross-dataset generalization. A smaller number of studies, exemplified by [27], use single-center, device-specific clinical acquisitions rather than public benchmarks, which supports deeper clinical

validation but constrains reproducibility and cross-study comparison. Overall, the dataset landscape reflects a field still anchored to a handful of legacy public benchmarks, with more recent work gradually extending coverage to multi-dataset and multimodal (OCT plus OCTA, or OCT plus fundus) evaluation.

RQ2: *What Kind of Techniques Are Used for Classification of OCT?*

Classification architectures across the reviewed corpus span the full progression from conventional convolutional networks to hybrid transformer-based designs. Standard pretrained CNN backbones, including VGG-family and ResNet-family architectures, remain heavily represented, appearing for instance in [29], [51], and [3]. Custom, task-specific architectures also recur, including the FCN-Nplus network of [1] and the Fully Dense Fusion CNN with Deep SVM and Deep KNN classifier heads of [56]. A distinct architectural trend, reflecting the field's broader shift toward global receptive-field modeling, is the adoption of transformer and hybrid CNN-Transformer designs, exemplified by the Class-Re-Attention Transformer of [37] and the dual-branch ResNet-50/Swin-Transformer fusion of [47]. Ensemble and fusion strategies form a further recurring pattern, seen in the weight-fused CNN ensemble of [58] and the multistage ResNet50/YOLOv3 cascade of [48]. Foundation-model fine-tuning represents the most recent development in the reviewed corpus, illustrated by the RETFound-OCT backbone fine-tuned in [27]. Across this range, classification architecture choice does not appear to be driven by explainability considerations; the same explainability techniques recur across conventional CNNs, hybrid transformers, and foundation models alike, suggesting that explainability has, to date, been treated largely as a technique bolted onto a classification architecture rather than as a co-designed component of it.

RQ3: *What Techniques Are Used in the Confidence of XAI?*

The reviewed corpus establishes confidence in its explanations through a comparatively narrow set of techniques, applied with widely varying rigor. Gradient-weighted Class Activation Mapping and its direct variants, principally Grad-CAM++ and Score-CAM, dominate the corpus, appearing in [1], [51], [52], and [36], among others, and are most frequently validated through qualitative visual inspection alone. Where quantitative confidence criteria are reported, three broad approaches recur. The first is inter-method similarity or agreement scoring, used by [1] and [51] to quantify the degree to which two CAM variants applied to the same model agree with one another. The second is ground-truth-grounded scoring, exemplified by the conformity metric of [44], which measures overlap between a Grad-CAM heatmap and expert-annotated lesion regions rather than agreement between two explanation methods. The third is formal statistical evaluation, most rigorously demonstrated

by the Insertion and Deletion analysis of [29] and the cross-metric correlation analysis of [3], the latter directly interrogating whether commonly used quantitative XAI metrics agree with one another at all. A smaller but methodologically distinctive strand of the corpus, represented by [27], establishes confidence in explanation output through structured independent expert clinical review rather than through an automatically computed metric. Techniques departing from the CAM family altogether, including the LIME-based approach of [56] and the SHAP-fusion approaches of [58] and [59], extend the confidence-building toolkit further but remain comparatively rare relative to CAM-centric approaches, and are rarely benchmarked against CAM-based confidence criteria on the same underlying model.

2.9 Conclusion

This chapter established the theoretical and technical foundation for the present research through a comprehensive study of retinal anatomy, retinal diseases, Optical Coherence Tomography, deep learning, and Explainable Artificial Intelligence, followed by a systematic review of the existing literature. The review demonstrates that while OCT-based retinal disease classification has reached a considerable level of maturity in terms of datasets, deep learning architectures, and diagnostic performance, explainability remains an active research challenge due to limited localization precision, inconsistent robustness, and the absence of standardized quantitative evaluation methods. These recurring limitations collectively justify the research gap identified in this dissertation and highlight the need for more reliable and clinically trustworthy interpretability frameworks. Building upon this baseline understanding, the following chapter presents a detailed comparative analysis of representative studies to examine existing explainability approaches, identify their strengths and limitations, and provide a stronger foundation for the proposed methodology.

CHAPTER 3

COMPARATIVE ANALYSIS OF XAI TECHNIQUES IN OCT

3.1 Introduction

The Systematic Literature Review conducted in Chapter 2 surveyed a broad corpus of studies concerned with Explainable Artificial Intelligence (XAI) as applied to Optical Coherence Tomography (OCT) and fundus-based retinal disease classification. That review, spanning close to forty peer-reviewed and preprint publications, established two things. First, the field has converged almost entirely on a small family of post-hoc attribution techniques, chiefly Gradient-weighted Class Activation Mapping (Grad-CAM) and its architectural variants, as the default mechanism for rendering deep classification networks interpretable. Second, despite this convergence, the manner in which these techniques are implemented, validated, and reported varies considerably from one study to the next, with only a minority of the reviewed corpus attempting any form of quantitative, ground-truth-grounded evaluation of the explanations produced.

This chapter narrows that broad literature base into a focused comparative analysis. Rather than re-describing the full set of reviewed studies, fifteen representative papers are selected from the larger pool and examined in depth, with particular attention paid to the specific explainability technique employed, the quantitative or qualitative evidence offered in support of that technique, and the limitations the original authors themselves acknowledge. The purpose of this narrower analysis is threefold: to surface the concrete methodological patterns that recur across the XAI literature applied to retinal imaging; to identify, through direct comparison, where these patterns converge and where they diverge; and to use the resulting synthesis as the evidentiary basis for the research problem and objectives that motivate the remainder of this dissertation.

The chapter proceeds as follows. Section 3.2 explains the basis on which the fifteen studies were selected and provides a consolidated overview table. Section 3.3 groups these studies by explainability technique family and discusses each group in turn, supported by academic tables extracted from the Systematic Literature Review data. Section 3.4 consolidates all fifteen studies into a single cross-cutting comparative table. Section 3.5 draws out the critical patterns emerging from this comparison and relates them back to the research problem and objectives established in Sections 1.3 and 1.6 of Chapter 1, and Section 3.6 closes the chapter by summarizing how the assembled evidence motivates the methodology presented in the following chapter.

3.2 Criteria for the Selection of Studies

The systematic literature review presented in Chapter 2 identified a broad body of research addressing Explainable Artificial Intelligence (XAI) for retinal disease

classification. While this body of literature provides a comprehensive understanding of the current state of research, conducting an in-depth comparative analysis of every identified study would reduce the clarity and focus of the discussion. Therefore, a representative subset of fifteen studies was selected for detailed comparative evaluation. The purpose of selecting this subset is not to identify the highest-performing studies, but rather to ensure adequate representation of the principal methodological directions observed throughout the literature. The selected studies collectively capture the diversity of explainability approaches, evaluation methodologies, experimental designs, and validation strategies currently employed in OCT-based retinal disease classification. This enables the comparative analysis to examine how different approaches address the common objective of improving the transparency and trustworthiness of deep learning models.

The selection process was guided by three principal criteria. First, methodological diversity was prioritized to ensure that the selected studies represented the major categories of explainability approaches identified during the systematic literature review. This allows the comparative analysis to evaluate a broad spectrum of techniques instead of repeatedly examining studies based on similar methodological principles.

Second, emphasis was placed on studies that provided meaningful evidence for evaluating explainability rather than reporting only classification performance. Preference was therefore given to research incorporating quantitative evaluation, expert validation, robustness assessment, localization analysis, or other forms of explanation validation that contribute to understanding the reliability of explainable AI methods.

Finally, domain relevance was considered essential throughout the selection process. Studies focusing on Optical Coherence Tomography and other clinically relevant retinal imaging modalities were prioritized to ensure that the comparative analysis remained directly aligned with the objectives of this dissertation and reflected the practical challenges encountered in ophthalmic image interpretation.

Applying these criteria resulted in a balanced set of fifteen representative studies that adequately capture the methodological diversity of the reviewed literature while maintaining a focused scope for detailed comparison. This selection provides an appropriate foundation for the comparative analysis presented in the following sections, where individual studies are examined with respect to their explainability techniques, evaluation methodologies, strengths, limitations, and overall contribution to trustworthy retinal disease classification.

Table 3.1 summarizes the fifteen selected studies, indicating the primary ex-

plainability technique employed, the imaging modality and backbone architecture used, and the technique family under which each study is discussed in Section 3.3.

Table 3.1: Overview of the Fifteen Studies Selected for Comparative Analysis

Ref.	Primary XAI Technique	Modality	Backbone Architecture	Ar-	Technique Family
[26]	Grad-CAM (training-integrated)	Fundus (ocular tumor)	VGG19 ResNet50 fusion	+	Standalone Grad-CAM
[47]	Grad-CAM (cross-dataset)	OCT	ResNet-50 Swin Transformer	+	Standalone Grad-CAM
[48]	Grad-CAM (vs. object detection)	OCT	ResNet50 YOLOv3	+	Standalone Grad-CAM
[1]	Grad-CAM & Score-CAM	OCT	FCNNplus		Comparative CAM benchmarking
[51]	Grad-CAM & Score-CAM	Angiography	StandardCNN, VGG16, MobileNet		Comparative CAM benchmarking
[44]	Grad-CAM (conformity metric)	Fundus	VGG/ResNet/DenseNet style		Comparative CAM benchmarking
[52]	Grad-CAM++	OCT	EfficientNetB0 + DeepLabV3+		Comparative CAM benchmarking
[36]	Grad-CAM & Grad-CAM++	Fundus	Attention U-Net + InceptionV3		Comparative CAM benchmarking
[55]	CAM-family (scoping review)	Fundus	Review scope		Comparative CAM benchmarking
[56]	LIME	OCT	Fully Dense Fusion CNN + SVM/KNN		Perturbation-based alternative
[29]	DeepLIFT, AGI, At-tEXplore	OCT / OCTA	VGG16		Perturbation/gradient alternative

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Table 3.1 continued from previous page

Ref.	Primary XAI Tech- nique	Modality	Backbone Ar- chitecture	Technique Family
[27]	RelProp / LRP	OCT	RETFound- OCT	Gradient- propagation alternative
[58]	Grad-CAM + SHAP	Fundus	Weight-fused CNN ensemble	Hybrid fusion
[59]	SCIM (SHAP-CAM fusion)	Fundus	VGG19	Hybrid fusion
[3]	Multi-method evalu- ation	Natural images	VGG16, ResNet-50	Evaluation co- herency

3.3 Comparative Analysis by Explainability Technique Family

Existing explainability methods for retinal disease classification can be broadly categorized according to the underlying principles used to generate visual explanations. Although these techniques share the common objective of improving model transparency, they differ considerably in methodology, interpretability, computational complexity, and practical applicability. Examining them according to their methodological families enables a more systematic comparison of their strengths and limitations.

3.3.1 Grad-CAM as a Standalone Explanation Mechanism

Three of the selected studies apply Grad-CAM as the sole explainability mechanism, without benchmarking it against an alternative CAM variant, and each does so in a way that illustrates a distinct strand of how standalone Grad-CAM is currently used in the literature.

The first strand treats Grad-CAM purely as a post-hoc diagnostic aid, generated after training is complete and never influencing the learning process itself. This is the conventional deployment pattern. A departure from it appears in the ocular tumor detection study of [26], which integrates Grad-CAM directly into the training loop through a heatmap-aligned loss function, using adaptive thresholding to derive proxy lesion masks in the absence of pixel-level ground truth. The resulting fusion architecture (combining VGG19 and ResNet50 feature extractors) demonstrated strong overall discriminative performance, which the authors attribute in part to the tighter coupling between the localization signal and the optimization objective. This is one of the few studies in the wider corpus where Grad-CAM output is not purely post-hoc, and it demonstrates that the localization behaviour of

Grad-CAM can, at least in principle, be shaped during training rather than merely inspected afterward.

The second strand uses standalone Grad-CAM as a validation device for a novel classification backbone rather than as the primary contribution. [47] proposes HyRetiNet, a dual-branch architecture fusing a ResNet-50 convolutional branch with a Swin Transformer global-attention branch, and applies Grad-CAM consistently across three separate public OCT benchmarks (OCT-2014, OCT-2017, and OCT-C8) to confirm that the fused network attends to class-discriminative lesion regions. Reported classification performance across the four-, three-, and eight-category tasks was consistently strong, making this one of the highest-performing architectures in the reviewed corpus; however, the authors explicitly note that no formal quantitative evaluation of Grad-CAM heatmap quality or faithfulness was performed on the hybrid architecture itself, meaning the explanation component is qualitatively illustrative rather than independently validated.

The third strand is more critical of standalone Grad-CAM. [48] constructs a multistage OCT screening pipeline (binary screening, followed by multi-label disease classification, followed by object-detection-based pathology localization using YOLOv3) and explicitly contrasts the interpretability richness of Grad-CAM heatmaps against explicit bounding-box localization. The classification stages achieved strong discriminative performance, and the object-detection stage achieved consistently high lesion-location recall across ten distinct lesion types. Critically, the authors state that reliance on Grad-CAM heatmaps alone "may not be as informative as the pathology detection approach," directly questioning whether CAM-based pixel-level attribution is the most clinically actionable form of explanation available, even though it remains the most widely adopted.

Taken together, these three studies show that standalone Grad-CAM is being pulled in three different directions within the same broad literature: toward tighter integration with the training objective, toward passive validation of ever more complex classification backbones, and toward outright substitution by object-detection-based localization. None of the three, however, offers a quantitative faithfulness metric for the Grad-CAM output itself, which is a limitation examined further in Section 3.3.2.

3.3.2 Comparative Evaluation of Multiple CAM Variants

A second and larger cluster of studies moves beyond standalone Grad-CAM to directly compare two or more CAM-family techniques on the same underlying model, or to propose a dedicated metric for quantifying CAM quality. This cluster represents the methodologically strongest segment of the reviewed corpus with

respect to explanation validation, and it is summarized in Table 3.2.

The researchers in [1] proposes FCNNplus, a novel CNN architecture for joint retinal disease classification and localization, and applies both Grad-CAM and Score-CAM to its output, introducing a similarity index to quantify agreement between the two CAM variants. The architecture demonstrated strong classification performance on the Kermany2018 four-class dataset, with somewhat reduced performance on the more heterogeneous Retinal OCT-C8 dataset, and the similarity index was reported for both. [51] performs a structurally similar comparison, applying Grad-CAM and Score-CAM across three backbones (StandardCNN, VGG16, and MobileNet) and two classification granularities (binary healthy-versus-DR and three-class DR staging), reporting a purpose-built similarity index for each model-CAM pairing rather than a single aggregate figure. Both studies converge on the same underlying observation: that CAM variants applied to the identical trained model can disagree with one another to a non-trivial degree, and that this disagreement is rarely reported elsewhere in the literature.

The researchers in [44] and [52] approach the comparative question from different angles. Rather than comparing two CAM variants against each other, [44] proposes a domain-knowledge "conformity" metric that measures the overlap between Grad-CAM heatmap activation and expert-annotated ground-truth lesion regions, offering a template for grounding CAM evaluation in clinical annotation rather than inter-method agreement alone. [52] instead compares Grad-CAM against its generalization, Grad-CAM++, within a segmentation-guided CAD pipeline (EfficientNetB0 classifier guided by DeepLabV3+ segmentation), reporting that Grad-CAM++ produced visibly clearer heatmaps than standard Grad-CAM on OCT imagery, with overall diagnostic performance improving once the segmentation-guided Grad-CAM++ pipeline was adopted. [36] reaches a broadly consistent conclusion in a two-stage glaucoma identification pipeline (Attention U-Net segmentation followed by InceptionV3 classification), applying Grad-CAM and Grad-CAM++ jointly rather than choosing between them, on the reasoning that the two variants offer complementary explanatory views of the same underlying decision.

Finally, the researchers in [55] synthesizes this entire comparative sub-literature at scale, reviewing twenty-eight studies across fourteen public diabetic retinopathy datasets and twelve distinct CAM-family variants (including Grad-CAM, Grad-CAM++, Score-CAM, Ablation-CAM, LayerCAM, Eigen-CAM, and others). The review identifies recurring, cross-study failure modes, including vessel-focused heatmaps, optic disc glare, and poor sensitivity to low-contrast microaneurysms, and concludes that causal faithfulness evaluation, as distinct from visual plausibility, re-

mains comparatively underdeveloped across the CAM literature as a whole.

Table 3.2: Quantitative Evidence Reported in Comparative CAM-Benchmarking Studies

Ref.	Comparative Mechanism	Principal Quantitative Finding
[1]	Grad-CAM vs. Score-CAM, novel similarity index	Similarity index reported across both benchmark datasets but not externally validated against clinical annotation
[51]	Grad-CAM vs. Score-CAM across three backbones	Similarity index varies by model-CAM pairing across both classification granularities
[44]	Grad-CAM vs. ground-truth lesion annotation	Conformity score quantifies heatmap-to-lesion overlap; metric not yet extended to OCT
[52]	Grad-CAM vs. Grad-CAM++	Diagnostic performance improved with Grad-CAM++ and segmentation guidance, alongside visibly clearer heatmaps than standard Grad-CAM
[36]	Grad-CAM and Grad-CAM++ jointly	CAM variants used as complementary rather than competing views of the same underlying decision
[55]	Meta-synthesis of 12 CAM variants across 28 studies	Recurring failure modes (vessel bias, disc glare, low microaneurysm sensitivity) identified at scale

3.3.3 Perturbation-Based and Gradient-Propagation Alternatives to CAM

A smaller but methodologically distinct set of studies departs from the CAM family altogether, employing perturbation-based, propagation-based, or otherwise model-agnostic attribution techniques. [56] explains a Fully Dense Fusion CNN, coupled with Deep SVM and Deep KNN classifier heads, using Local Interpretable Model-agnostic Explanations (LIME) rather than any gradient-based CAM technique. LIME generates superpixel-based, perturbation-derived explanations that do not require access to internal network gradients, and the reported classification performance was strong across both benchmark datasets evaluated; however, the authors themselves note that no comparative evaluation against Grad-CAM or its variants was performed on the same architecture, leaving open the question of whether LIME’s superpixel explanations are more, less, or comparably faithful to the network’s true decision logic.

The researchers in [29] takes the comparative question that [56] leaves open and

applies it directly, benchmarking three attribution methods, DeepLIFT, Adversarial Gradient Integration (AGI), and AttEXplore, within a single VGG16 pipeline evaluated on both the Kermany OCT dataset and widefield OCTA imagery, using quantitative Insertion and Deletion scores. The reported scores (for example, Insertion/Deletion of 0.828/0.740 for DeepLIFT and 0.938/0.680 for AttEXplore on OCTA) reveal that different attribution methods applied to an identical trained model produce measurably different, and at times contradictory, rankings of explanation quality, echoing the inter-CAM disagreement already observed in Section 3.3.2 but for a different family of techniques entirely.

the researchers in [27] represents a third and clinically distinctive strand, applying Relevance Propagation (RelProp / Layer-wise Relevance Propagation) to a fine-tuned RETFound-OCT foundation model in order to identify neuroretinal structural biomarkers of diabetic retinal neurodegeneration. Rather than benchmarking against classification accuracy alone, this study grounds its explanation output in structured clinical review: two independent retinal experts evaluated the resulting relevance maps against four specific anatomical signs, reporting inter-rater agreement (Cohen's kappa) ranging from 0.88 to 0.98 across signs such as ellipsoid zone disruption and choroidal hypertransmission. This is one of the few studies among the fifteen selected here to validate its explanation output through structured expert clinical review rather than through an automatically computed image-similarity or overlap metric, and its use of an OCT-native foundation model (RETFound-OCT) further distinguishes it from the ImageNet-pretrained CNN backbones used elsewhere in the corpus.

3.3.4 Hybrid Fusion of CAM and Feature-Attribution Methods

Two of the selected studies pursue explicit hybridization between spatial, CAM-based localization and feature-level, game-theoretic attribution, on the reasoning that the two approaches answer complementary questions: CAM indicates *where* in the image the network is attending, while SHAP-style attribution indicates *which learned features* most influenced the prediction. [58] combines Grad-CAM with SHapley Additive exPlanations (SHAP) within a weight-fused ensemble of pre-trained CNNs for diabetic retinopathy grading, evaluated across three independent fundus datasets (APTOS, MESSIDOR, IDRiD), with the ensemble demonstrating strong overall discriminative performance across all three. [59] goes further, proposing SCIM (SHAP-CAM Interpretable Mapping), a genuine algorithmic fusion of SHAP attribution values with CAM-based spatial localization into a single interpretable map for glaucoma classification using a VGG19 backbone, and directly benchmarks the fused technique against standalone Grad-CAM and stan-

alone SHAP.

Both studies report that the fused explanation is qualitatively more informative than either constituent technique used in isolation, yet neither offers a formal, quantitative agreement or consistency score between the spatial and feature-level components being fused. This is a specific and recurring gap: the hybridization intuition, that spatial and feature-level attribution are complementary rather than redundant, is plausible and consistently asserted across both studies, but it has not yet been quantitatively demonstrated in the reviewed literature, nor has either fusion approach been extended to OCT cross-sectional imagery, where the anatomical layer structure absent from fundus photography could plausibly change how spatial and feature-level explanations relate to one another.

3.3.5 Quantitative Evaluation Coherency Across Explanation Methods

The final study in this synthesis addresses a question that cuts across every technique family discussed above: given that so many of the preceding studies report a single quantitative metric (a similarity index, a conformity score, an Insertion/Deletion pair) as evidence of explanation quality, how reliable are these metrics themselves? [3] investigates this directly, comparing eight distinct explanation methods, including Grad-CAM, Integrated Gradients, SmoothGrad, LRP, sliding-window occlusion, RISE, adaSISE, and TAME, across two backbone architectures (VGG16 and ResNet-50) on ImageNet, using a battery of quantitative evaluation metrics (Insertion/Deletion AUC, Average Drop, Increase-in-Confidence, ROAD, and the pointing game) together with formal correlation analysis between metrics.

The central finding is that these metrics are not mutually consistent: Insertion and Deletion scores were found to correlate strongly and negatively with one another (Spearman's $\rho \approx -0.7$), while Deletion and ROAD correlated strongly and positively ($\rho \approx 0.9$), indicating that some metric pairs encode genuinely opposing notions of what makes an explanation "good," while others are functionally redundant. Although this study is conducted on natural-image classification rather than retinal imagery, its implications apply directly to every quantitative metric introduced elsewhere in this chapter, including the similarity indices of [1] and [51] and the conformity score of [44]: without an independent coherency check of the kind [3] performs, a single reported metric value cannot, by itself, establish that one explanation is more faithful than another.

3.4 Cross-Cutting Comparative Table

Table 3.3 consolidates all fifteen studies discussed in Section 3.3 into a single comparative view, indicating for each study the reported quantitative validation of

its explainability output, if any, and the principal limitation acknowledged by the original authors.

Table 3.3: Cross-Cutting Comparison of the Fifteen Selected Studies

Ref.	Reported Quantitative Explanation Validation	Principal Acknowledged Limitation
[26]	None; Grad-CAM embedded in training loss but not independently scored	No pixel-level ground-truth annotation available
[47]	None; Grad-CAM shown qualitatively only	No formal faithfulness evaluation of Grad-CAM on the hybrid architecture
[48]	Indirect, via qualitative comparison with explicit object-detection-based localization	Grad-CAM heatmaps judged less informative than explicit localization
[1]	Similarity index (Grad-CAM vs. Score-CAM)	Similarity index not externally validated
[51]	Similarity index across three backbones	Index measures inter-CAM agreement, not agreement with ground truth
[44]	Conformity score vs. lesion annotation	Requires costly expert lesion-level annotation; not yet applied to OCT
[52]	Qualitative comparison, Grad-CAM vs. Grad-CAM++	Evaluated on a subset of a single public dataset
[36]	Qualitative, two CAM variants jointly	Limited dataset diversity, no external validation
[55]	Meta-level synthesis (IoU, Dice, DAUC/IAUC, ADCC)	Causal faithfulness evaluation underdeveloped across reviewed studies
[56]	None; LIME used without CAM comparison	No comparative evaluation against Grad-CAM on the same model
[29]	Insertion/Deletion scores across three methods	Methods yield divergent spatial localizations despite operating on an identical underlying model
[27]	Expert inter-rater agreement ($\kappa = 0.88\text{--}0.98$)	Single-center design; specific device and OCT section type
[58]	None; Grad-CAM and SHAP combined qualitatively	No agreement metric between Grad-CAM and SHAP outputs

Continued on next page

Table 3.3 continued from previous page

Ref.	Reported Quantitative Explanation Validation	Principal Acknowledged Limitation
[59]	Qualitative comparison, SCIM vs. Grad-CAM vs. SHAP	No large-scale quantitative validation of the fusion technique
[3]	Formal correlation analysis across eight methods	Restricted to natural images and two architectures

3.5 Critical Discussion

The comparative analysis demonstrates that considerable progress has been achieved in applying Explainable Artificial Intelligence to retinal disease classification; however, it also reveals several recurring limitations that extend across the majority of the selected studies. Although the reviewed works differ in their datasets, network architectures, and explainability techniques, they exhibit common methodological trends that collectively define the current state of the field. Rather than representing isolated shortcomings of individual studies, these recurring observations establish broader research challenges that continue to limit the clinical reliability of explainable diagnostic systems.

One of the most consistent observations concerns the generation of explanations themselves. Across the selected studies, Grad-CAM and its variants remain the dominant explanation mechanism because of their simplicity and compatibility with convolutional neural networks. Despite their widespread adoption, most implementations continue to derive explanations primarily from the final convolutional layer, producing heatmaps with limited spatial resolution due to repeated downsampling during feature extraction. Several studies acknowledge this limitation either by proposing modified CAM variants or by combining information from multiple explainability methods; however, no generally accepted solution has yet emerged that consistently produces high-resolution, clinically meaningful localization across different retinal pathologies [1, 47, 51, 52]. This recurring dependence on coarse last-layer explanations directly supports the first research gap identified in this dissertation.

The comparative analysis further reveals that explanation validation remains significantly less mature than classification evaluation. Nearly every study reports comprehensive classification metrics, including accuracy, sensitivity, specificity, precision, and AUC, whereas considerably fewer provide quantitative evidence demonstrating the reliability of the generated explanations. Where validation is performed, the approaches differ substantially, ranging from expert assessment and localization

analysis to similarity indices, perturbation experiments, and insertion-deletion metrics [3,27,29,44]. The absence of a standardized validation framework makes direct comparison between studies difficult and limits confidence in the reported explanations. Consequently, the literature continues to emphasize predictive performance more strongly than explanation quality, reinforcing the second research gap concerning the lack of consistent quantitative evaluation.

A related observation concerns the robustness of explainability methods. Only a limited number of studies explicitly investigate whether explanations remain stable under varying imaging conditions or input perturbations. Most evaluations are conducted under ideal experimental settings, providing relatively little evidence regarding the consistency of explanations when images are affected by noise, acquisition variability, or other practical clinical conditions. Since robustness is fundamental to trustworthy clinical decision support, this limited attention represents another important gap that remains insufficiently addressed within existing research.

The comparison also highlights that, although numerous explainability techniques have been proposed, relatively few studies attempt to integrate accurate classification, reliable explanation generation, and rigorous explanation validation within a unified framework. Instead, improvements are commonly directed toward one aspect of the problem while assuming that the remaining components are adequate. This fragmented development has resulted in explainability methods that are often visually convincing but only partially validated for practical clinical use. Consequently, despite the substantial advances made in automated retinal disease classification, the objective of producing explanations that are simultaneously accurate, robust, quantitatively validated, and clinically trustworthy remains largely unresolved.

Collectively, these findings strengthen the rationale established in the preceding chapters and provide direct evidence supporting the research objectives of the present study. The literature review established the theoretical foundations of OCT imaging and explainable artificial intelligence, while the comparative analysis has identified the practical limitations that continue to characterize current explainability research. These observations justify the need for a framework capable of improving localization precision, strengthening explanation robustness, and incorporating a more systematic strategy for quantitative validation. The methodology presented in the following chapter is therefore designed to address these limitations by integrating robust pixel-level interpretability with an evaluation framework intended to provide greater confidence in the generated explanations.

3.6 Conclusion

The comparative analysis presented in this chapter provided a detailed examination of representative studies in OCT-based retinal disease classification and explainable artificial intelligence. Although significant progress has been achieved in developing highly accurate deep learning models, the analysis demonstrates that important challenges remain in producing explanations that are precise, robust, quantitatively validated, and clinically reliable. These observations reinforce the research gaps identified in the earlier chapters and confirm that existing approaches have not yet fully addressed the requirements for trustworthy clinical decision support. Consequently, the findings of this chapter provide a strong foundation for the objectives of the present study by identifying the methodological limitations that require further investigation. Building upon the theoretical foundations established through the background study, the systematic literature review, and the comparative analysis, there is a need to design a robust model for explaining OCT classification, which covers the challenges discussed in the previous chapters.

CHAPTER 4

A ROBUST PIXEL-LEVEL INTERPRETABILITY IN OCT-BASED RETINAL DISEASE CLASSIFICATION

4.1 Introduction

Retina is a light-sensitive neural tissue layer at the back of eye consisting of photoreceptor cells, namely rods and cones. It converts the incoming light into neural signals, which are then sent to the brain enabling vision. [38]. Retinal diseases are a major cause of visual impairment around the world and can result in irreversible blindness in severe cases. Among these, Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME) and Drusen are the major pathological conditions. CNV refers to the abnormal growth of blood vessels arising from the choroid into retinal layers [25]. DME arises due to Hyperglycemia causing fluid in the retinal blood vessels to leak in the macula [25]. Drusen is caused by the accumulation of extracellular proteins and lipids between the Bruch's membrane and the basal lamina of the Retinal Pigment Epithelium (RPE) [20]. To critically examine the retinal tissues, there is a need for non-invasive medical techniques. Optical Coherence Tomography (OCT) addresses this, becoming a gold-standard for retinal imaging diagnosis [6]. OCT is a non-contact and high-resolution optical technique for micrometer-scale cross-sectional visualization of retinal structures [24], providing a detailed image of retinal structures.

In Deep Learning (DL), Convolutional Neural Network (CNN) has been the most efficient technique for extracting reliable retinal features effectively by utilizing OCT images [24]. In recent implementations, multiple CNN-based models have been evaluated on different public and private datasets achieving classification accuracies ranging from 96.6% to 99.97%, successfully [25] [62] [32]. Even after achieving high classification accuracy, CNNs are always susceptible due to their "black-box" nature, raising issues with trust and confidence. The Ophthalmologists fail to grasp the reasoning of the model responsible for the prediction and diagnosis of a specific condition from an OCT image input [1]. To make the predictions transparent and reduce bias, black-box CNN models are transformed into Explainable Artificial Intelligence (XAI) models [25] [1]. Class Activation Mapping (CAM) based methods are mostly used for the explainability of CNN models. They highlight the specific areas of an input image which primarily influence the model's decision making, by generating heatmaps over them. [25] [1] Out of multiple CAM based techniques, Gradient Class Activation Map (Grad-CAM) turns out to be one of the most important and widely used explainability techniques for CNNs, particularly in retinal image classification models. [32] [1] Results show, Grad-CAM has achieved a high Dice coefficient of 78.62%, outperforming other baseline models for explanation [63]. While Grad-CAM has been extensively applied for interpretability in CNN-based disease diagnosis, it faces some shortcom-

ing, such as coarse spatial resolution, resulting in blurry and granular localization of lesions. [3, 35, 37, 55]. In addition, it does not provide precise localization of boundaries and lacks comprehensive validation approach affecting the validity of the explanation. [3, 55, 63] [28].

4.2 Literature Review

The researchers have utilized various public and private datasets, consisting of multiple retinal diseases, for detection and classification tasks. The publically available multi-class dataset OCT2017 consisting of 84484 OCT B-scans (CNV, DME, DRUSEN and NORMAL), collected by [38], serves as a gold standard for CNN based retinal disease classification [38] [32] [1]. The OCT-C8 dataset consists of 24000 OCT images that are divided into eight different classes has also been used by multiple researchers for evaluation of CNNs over the wide range retinopathic diseases [25] [32]. The custom datasets collected and evaluated by other researchers include Duke Univeristy datasets, a custom dataset collected from Shifa Eye Foundation Hospital Pakistan consisting of 7314 OCT images, Retinal Fundus Multi-Disease image Dataset (RFMiD) consisting of 3200 fundus images, a private dataset comprising of 1437 color fundus images and private Masked Trajectory Model (MTM) dataset consisting of five classes [38] [63] [64] [4].

To effectively make use of the OCT-based datasets in medical diagnosis, researchers have used a wide variety of standard CNN architectures. The VG19 architecture was evaluated on Shifa Eye Foundation Dataset [38]. The particular model achieved an accuracy of 97.54% , with corresponding precision, recall, and F1 scores above 97% . However, the SMOTE oversampling did not reflect real pathology [38]. Similarly, using a dataset of 1,437 color fundus images, the potential of different pre-trained CNNs [40], was explored resulting in the following accuracy results: Xception 89.27%, VGG16 91.44%, InceptionV3 89.06% and DenseNet121 93.35% . The proposed model addressed only binary DR stage classification (NPDR vs PDR) [40] . Furthermore, a 38-layer ResNet architecture with 37 convolution layers and 17 residual networks, increased the number of layers preventing the gradient loss. [2]. The particular model achieved moderate predictive correlation of 0.47. The manual 200×200 cropping however introduces operator-dependent bias and Grad-CAM heatmap was selectivity is qualitatively defined [2].

Beyond standard architectures, researchers have proposed custom and hybrid frameworks to improve classification performance. A lightweight retinal OCT image classification model integrating CNN and Transformer was proposed with few parameters [32]. The best accuracy of 98% and recall of 97.99% was achieved on the OCT-C8 dataset. The dataset heterogeneity may influence model performance

and no explainability techniques were applied to visualize model decisions [32]. A 2D single-channel CNN using ResNet101-V2 was built to leverage the capabilities of DNNs for detecting eye diseases [25], achieving an accuracy of 99.90%, with precision 99.80%, recall 99.79%, and F1-score of 99.79%. In this architecture Grad-CAM heatmaps were assessed only qualitatively, alongside 2D single-channel input may not fully exploit color information [25]. A deep learning model, FCN-Nplus, was proposed and designed for the classification of retinal disease [1], where accuracy of 93.3% was achieved. In this model the achieved accuracy (93.3%) was lower than other transfer learning models alongside the CAM heatmaps lack fine-grained localization [1].

To address the black-box problem of deep learning models and to enhance their transparency and interpretability, the researchers have proposed Explainable-AI (XAI) as a solution. [27]. In this regard, researchers have identified visual explanation methods which assign scores to features in the input image based on their relevance, and numerous explanation methods have been introduced [55] [3]. Among these methods, researchers have noted that Grad-CAM, remains the most widely used CAM-based method [3]. Numerous studies have incorporated Grad-CAM into their retinal disease classification pipelines to enhance clinical interpretability [25] [28]. The incorporation of Grad-CAM into the CNN framework for the clinical interpretability showed the results, that the most important regions responsible for the predicted class were the highlighted ones. [28]. Grad-CAM visualizations have highlighted the regions which were anatomically and clinically meaningful, including areas where diabetic retinopathy lesions are known to occur like the areas around the macula, vascular structures and optic disc [28]. The study presented in Detection of Central Retinal Artery Occlusion (CRAO) using OCT, observed that regarding the pathological and morphological changes to mentioned class and the differential diagnosis class, the focus of the model was on the changes in middle retinal layer specifically the macular periphery. [35]. Collectively, these studies demonstrate the significant clinical value of Grad-CAM in retinal disease analysis. Despite being the most commonly used explainability technique, several technical limitations of Grad-CAM have been identified that reduce its effectiveness in retinal disease classification.

The major limitation involves low spatial resolution of Grad-CAM outputs. Both CAM and Grad-CAM typically produce low-resolution and coarse heatmaps that blur lesion boundaries [55]. This occurs because of the gradients naturally being sparse in their nature, while the up-sampling of very small feature maps further reduces spatial precision [3]. In addition, traditional Class-activation mapping

methods suffer from either poor localization of features due to low spatial resolution or immense noise arising from techniques based on gradients. This limitation primarily stems from their reliance on the final convolutional layer [55]. Comparative studies have further revealed that the CNN methods focus on regions largely effected by the disease, while the Vision Transformer (ViT) methods are more focused towards the refined disease regions. This suggests the CNN-based Grad-CAM explanations are inherently limited in granularity, which contribute to the resolution challenge. [37] [35]. Since the retinal diseases require precise localization of tissues and regions [55], the attribution of generated heatmaps seems to be very unrealistic, because they fundamentally reflect the model's prediction process rather than precise pixel segmentation, which fails to guarantee the pixel-level accuracy [29] [27]. therefore, Grad-CAM localization when intergrated with such architectures introduce spatial noise which may not perfectly represent the true lesion regions, reducing the localization precison [63] [27] [26].

4.3 PROPOSED METHODOLOGY

This study proposes a Multi-Layer Fused Grad-CAM, while standard Grad-CAM only looks at one convolutional layer, and that single view loses a lot of useful spatial information. The goal here is to pull activation maps from several network depths, weigh them according to how much diagnostic signal each actually carries, and fuse them into a single, more informative heatmap. Robustness under noise is then used as a concrete way to test whether the fused explanations are genuinely more stable than the conventional baseline.

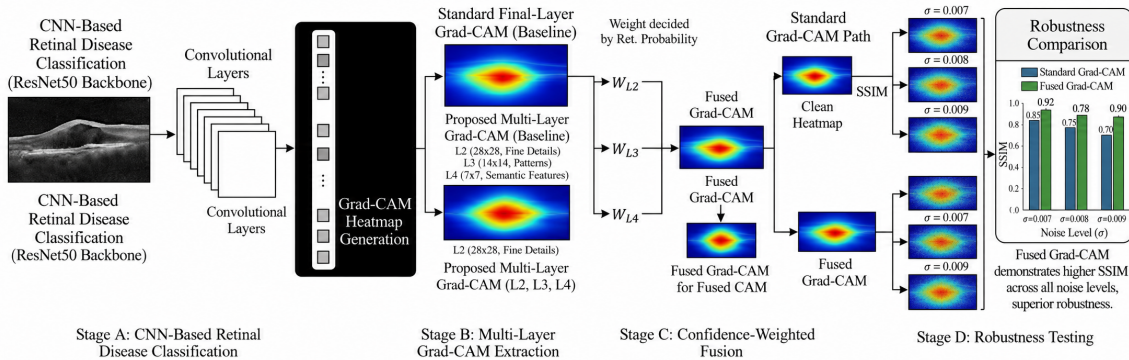


Figure 4.1: Four Stage Multi-Layer Fused Grad-CAM framework.

4.3.1 Dataset

The classification model was tested on the publicly available OCT2017 dataset introduced by Kermany [32]. OCT2017 is one of the most widely adopted benchmark datasets for automated retinal disease classification [32] [40]. The dataset

contains retinal OCT B-scan images comprising four classes : Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), Drusen, and Normal retina. The available distribution of the dataset is presented in Table 4.1.

Table 4.1: Class distribution of the OCT2017 dataset.

Class	Training	Validation	Testing	Total
CNV	37,205	8	242	37,455
DME	11,384	8	242	11,634
DRUSEN	8,616	8	242	8,866
NORMAL	26,315	8	242	26,565
Total	83,484	32	968	84,484

4.3.2 CNN-Based Retinal Disease Classification

The OCT2017 dataset consists of four categories: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), DRUSEN, and NORMAL. Before training, all OCT B-scans were resized to 224×224 pixels and normalized to a consistent range.

A ResNet50 pre-trained on ImageNet was used as the backbone. The trained model reached roughly 98% overall accuracy, which is stable enough to treat the network's decisions as trustworthy inputs for the explainability stage.

4.3.3 Standard Grad-CAM Generation

Standard Grad-CAM was applied to the final convolutional layer as the baseline explanation method. For a target class c , layer importance weights are the average gradient of the class score y^c with respect to each feature map A^k :

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (4.1)$$

These weights further form a weighted sum of the feature maps, which is passed through a ReLU to keep only positive activations:

$$L_{GradCAM}^c = ReLU \left(\sum_k \alpha_k^c A^k \right) \quad (4.2)$$

The resulting heatmap marks which image regions most influenced the network's decision for class c .

4.3.4 Multi-Layer Fused Grad-CAM

Rather than stopping at the final layer, the proposed method extracts Grad-CAM heatmaps from three intermediate convolutional layers. Each layer captures a different level of detail:

- Layer 2 (28×28): fine spatial details,
- Layer 3 (14×14): mid-level structural patterns,
- Layer 4 (7×7): high-level semantic features.

Running Grad-CAM separately at each depth produces three layer-specific activation maps, L_2 , L_3 , and L_4 .

4.3.5 Confidence-Based Fusion

The three maps need to be combined, but a simple equal-weight average ignores the fact that some layers will be more informative than others depending on the image. The confidence-retention mechanism decides the weights empirically, per image.

Each layer's heatmap is first binarized into a mask and applied to the original OCT scan, blocking out all regions the layer did not highlight. The masked image is then passed back through the trained CNN, and the retained class probability is recorded:

$$r_l = P(c|I_{masked}) \quad (4.3)$$

The idea is direct: if masking a layer's heatmap still leaves the classifier confident, that layer was actually capturing relevant features. The weight for each layer is its retention score normalized across all three layers:

$$w_l = \frac{r_l}{\sum_l r_l} \quad (4.4)$$

The final fused heatmap is then a weighted sum of the three layer maps:

$$L_{fused} = w_2 L_2 + w_3 L_3 + w_4 L_4 \quad (4.5)$$

Layers that preserve classifier confidence get more weight; layers whose masked images confuse the classifier contribute less. This keeps the fusion grounded in what the model actually uses, rather than applying uniform weights regardless of what each layer captures.

4.3.6 Robustness Evaluation Using SSIM

A practical explanation method should produce consistent heatmaps when the input has minor variations. To test this, Gaussian noise was added to both the standard and fused heatmaps:

$$H' = H + N(0, \sigma^2) \quad (4.6)$$

where $N(0, \sigma^2)$ is Gaussian noise with variance σ^2 . Three noise levels were tested ($\sigma = 0.007, 0.008, 0.009$), covering a range from low perturbation to more aggressive corruption.

The Structural Similarity Index Measure (SSIM) was used to quantify how much each heatmap degraded under noise:

$$SSIM(H, H') \quad (4.7)$$

A score closer to 1 means the clean and noisy heatmaps are nearly identical, i.e., the explanation is stable. Adaptive Fused Grad-CAM and Standard Grad-CAM were compared at all three noise levels, across all four disease classes.

4.4 RESULTS AND DISCUSSION

The Multi-Layer Fused Grad-CAM framework was tested on the OCT2017 dataset, covering CNV, DME, DRUSEN, and NORMAL.

Qualitative inspection of the generated heatmaps in Standard Grad-CAM as shown in Fig 4.2a, constrained to the final convolutional layer, tended to produce broad, diffuse activations that spread across the scan without tightly following the lesion boundary. The multi-layer fused maps were noticeably more localized as shown in Fig. 4.2b: the activated regions corresponded more closely to the structural features that distinguish each disease class, which makes sense given that the shallower layers contribute spatial detail that the deepest layer alone cannot recover.

For a quantitative comparison, Gaussian noise was injected into both sets of heatmaps and SSIM was measured between the clean and noisy versions. The results are in Table 4.2.

The Multi-Layer Fused Grad-CAM outperformed Standard Grad-CAM at every noise level. The fact that the gap grows with noise level is the more interesting result: it means fused heatmaps do not just start from a slightly higher baseline, they degrade more slowly as conditions worsen. This is consistent with the multi-scale design. Shallow layers provide fine-grained spatial detail; if one scale gets corrupted, the others partially compensate, and the confidence-retention weighting automatically down-weights any layer whose activation structure has become in-

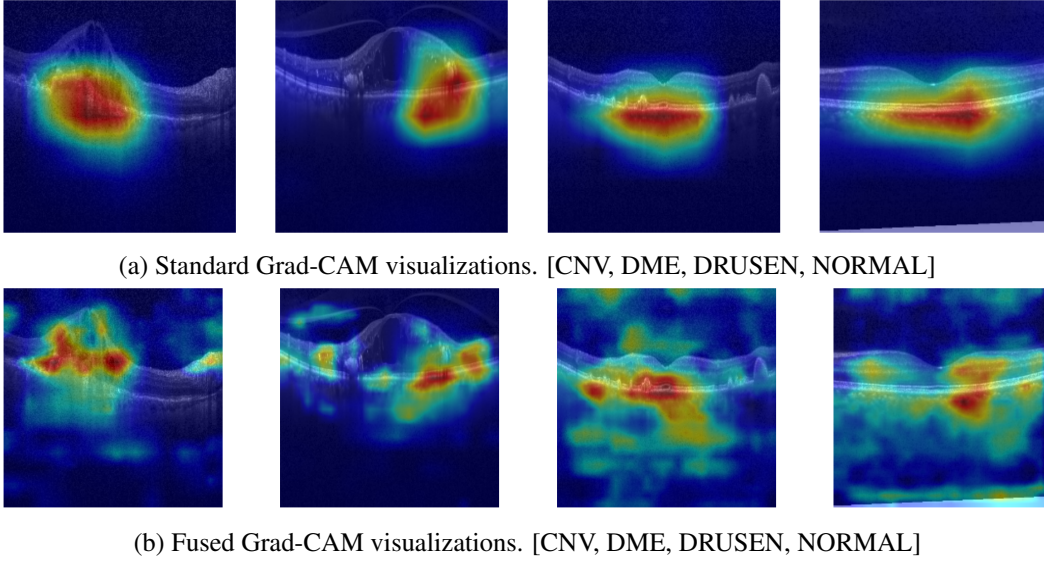


Figure 4.2: Heatmap Comparison.

Table 4.2: Class-wise SSIM robustness under Gaussian noise.

σ	Class	Fused	Standard	Gain
0.007	CNV	0.9713	0.9543	0.0170
	DME	0.9698	0.9502	0.0196
	DRUSEN	0.9750	0.9566	0.0184
	NORMAL	0.9736	0.9579	0.0157
0.008	CNV	0.9630	0.9413	0.0217
	DME	0.9613	0.9349	0.0264
	DRUSEN	0.9674	0.9445	0.0229
	NORMAL	0.9655	0.9470	0.0185
0.009	CNV	0.9532	0.9274	0.0258
	DME	0.9516	0.9195	0.0321
	DRUSEN	0.9592	0.9313	0.0279
	NORMAL	0.9570	0.9337	0.0233

consistent.

The observations in Table 4.2 support the central claim: integrating multi-scale feature representations through evidence-based weighting produces explanations that are both more anatomically focused and more stable under realistic degradation conditions.

4.5 CONCLUSION

This work proposed a Multi-Layer Fused Grad-CAM framework for explainable retinal disease classification from OCT images. Instead of generating a saliency map from a single convolutional layer, the method draws from three network depths and uses confidence-retention scores to set each layer's contribution prior to their fusion into a single heatmap. Robustness testing, conducted by adding Gaussian noise at three levels and measuring SSIM against clean heatmaps, showed that the fused maps consistently outperformed standard Grad-CAM. The performance gap also grew with increasing noise, which suggests the multi-scale fusion provides some genuine structural redundancy rather than just a flat improvement in baseline quality.

Despite this, obvious limitations remain. The evaluation uses SSIM under synthetic noise as a proxy for robustness, which doesn't fully capture the variability seen in real-world OCT acquisitions across different scanners and imaging protocols. Future work should incorporate expert-annotated lesion masks to assess whether the activated regions correspond to clinically defined pathology. Extending the approach to other CNN backbones and to OCT-A data would also clarify how broadly the method generalizes.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

Reliable explainability remains one of the main barriers preventing deep learning models from being confidently adopted in clinical ophthalmology. Although modern convolutional neural networks achieve high classification accuracy on Optical Coherence Tomography (OCT) images, clinicians still require evidence that the regions highlighted by explainability methods genuinely reflect the pathological features influencing a prediction. This work was motivated by the observation that widely used methods such as Grad-CAM typically rely only on the final convolutional layer, producing coarse localization maps whose reliability is rarely evaluated beyond visual inspection.

The work began by establishing the clinical and technical background of OCT imaging before conducting a Systematic Literature Review of explainable artificial intelligence techniques for retinal disease classification. The review showed that research has progressed rapidly in improving classification performance, whereas comparatively little attention has been given to validating the trustworthiness of the explanations produced by these models. Most studies continued to rely on qualitative comparisons, with only a limited number employing quantitative evaluation metrics.

A detailed comparative analysis of representative studies further highlighted the strengths and limitations of the major explainability approaches. While several CAM-based, perturbation-based, and hybrid techniques have been proposed, no single method consistently demonstrated superior performance across different datasets and evaluation criteria. More importantly, existing confidence metrics often measured different aspects of explanation quality, making meaningful comparisons between studies difficult. These observations confirmed the need for an explainability framework that not only improves localization but can also be evaluated using objective measures.

To address this gap, a Multi-Layer Fused Grad-CAM framework was developed using a ResNet50 backbone. Instead of generating explanations solely from the final convolutional layer, the proposed approach combines activation maps extracted from multiple intermediate layers through an adaptive confidence-retention weighting strategy. This allows fine spatial details from earlier layers to complement the stronger semantic information available in deeper layers. Experimental evaluation under progressively increasing Gaussian noise demonstrated that the fused approach consistently preserved structural information better than conventional Grad-CAM, as measured using the Structural Similarity Index Measure (SSIM). The improvement became more pronounced as noise levels increased, indicating that multi-layer

fusion provides greater robustness rather than merely improving visual appearance under ideal conditions.

5.2 Future Work

Although the proposed framework demonstrates improved robustness and finer pixel-level interpretability compared with conventional Grad-CAM, several opportunities remain for future research. The most significant limitation of the present work is the absence of a publicly available expert-annotated OCT dataset containing precise lesion-level ground truth for explainability evaluation. Consequently, the generated heatmaps were assessed using quantitative image-similarity measures rather than direct comparison with clinically annotated pathological regions. A major direction for future work is therefore to collaborate with experienced retinal specialists to develop a comprehensive pixel-level annotated OCT dataset that identifies clinically relevant biomarkers such as intraretinal fluid, subretinal fluid, drusen, and choroidal neovascularization. Such a dataset would enable objective evaluation of explainability methods through direct comparison between generated explanations and expert-defined pathological regions, providing stronger clinical evidence for the reliability of the proposed framework.

Future research should also investigate the performance of the proposed framework using multi-centre datasets acquired from different OCT scanners and clinical environments. Evaluating the methodology across diverse imaging devices and acquisition protocols would provide a more comprehensive assessment of its robustness and generalizability under real-world clinical conditions.

Finally, although the proposed adaptive multi-layer fused Grad-CAM framework was developed using a ResNet50 backbone for two-dimensional OCT B-scan classification, its applicability to other deep learning architectures remains an open research direction. Extending the framework to modern convolutional networks, which will further establish its versatility and clinical applicability across a broader range of ophthalmic diagnostic systems.

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